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Pseudoperfect Numbers and the Jaccard Distance of Pseudoperfect Sets

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1 Introduction

Perfect numbers are a well-studied topic in the realm of Number Theory dating back millennia. In plain language, a perfect number is a number for which the sum of the proper divisors of the number equals the number. Euclid famously discovered 6, 28, 496, and 8128 to be the first four perfect numbers. For instance, the number 6 can be written as:

$$6 = 1 + 2 + 3$$

as 1, 2, and 3 are the proper divisors of 6. Similarly, for the number 28, we can write

$$28 = 1 + 2 + 4 + 7 + 14$$

where 1, 2, 4, 7, and 14 are the proper divisors of 28. Far more numbers have been established as perfect numbers since. Yet, the existence of infinitely many perfect numbers remain an open question. A current method for constructing perfect numbers is

$$m = 2^{n-1}\left(2^n - 1\right),$$

where $2^n - 1$ is prime. It can be checked that the proper divisors of m sum to m. The examples mentioned above can indeed be expressed in this form:

$$\begin{aligned} 6 &= 2^1\left(2^2 - 1\right), \\ 28 &= 2^2\left(2^3 - 1\right), \\ 496 &= 2^4\left(2^5 - 1\right), \text{ and} \\ 8128 &= 2^6\left(2^7 - 1\right). \end{aligned}$$

Prime numbers of the form $2^n - 1$ for an integer n are in fact well-known as *Mersenne primes*, and much like perfect numbers, the existence of Mersenne primes remains open.

More recently, in 1965, Wacław Sierpiński introduced the notion of *pseudoperfect numbers*, also referenced as *semi-perfect numbers*. These are numbers for which some subset of the proper divisors sum to the original number, and therefore the pseudoperfect numbers are a subset of the perfect numbers.

In our project, we wished to investigate the pseudoperfect numbers further. In particular, we dive into *primitive pseudoperfect* and *pseudoperfect sets*. For primitive pseudoperfect numbers, we establish an upper bound on the expression $\frac{\sigma(n)}{n}$ for a subset of primitive pseudoperfect n and formulate ways to construct specific primitive pseudoperfect aside from the well-known construction of $2^k \cdot p$ for a prime p and integer k satisfying $2^k < p < 2^{k+1}$. Within pseudoperfect sets, we prove there exists infinitely many pseudoperfect numbers with a fixed number of disjoint pseudoperfect sets for any number of sets, as well as establish a method to compare the elements of pseudoperfect sets and prove several properties of the method.

2 Background on Pseudoperfect Numbers

Definition 1 *For a positive integer n, we say $\sigma(n)$ is the sum of positive integer divisors of n. That is,*

$$\sigma(n) = \sum_{d|n} d.$$

For instance, $\sigma(6) = 1 + 2 + 3 + 6 = 12$ and $\sigma(7) = 1 + 7 = 8$.

Definition 2 *For a positive integer n, we say the d is a **proper divisor** of n if $d \mid n$ and $d \neq n$. In other words, the set of proper divisors of n is all divisors of n excluding itself.*

Definition 3 *A positive integer n is a **perfect number** if*

$$n = \sum_{\substack{d \neq n \\ d \mid n}} d$$

Equivalently, $\sigma(n) = 2n$ or, preferably, $\frac{\sigma(n)}{n} = 2$.

The discussion of perfect numbers which satisfy $\sigma(n) = 2n$ leads us to explore numbers where $\sigma(n) \neq 2n$. Thus, leading us to the following definition.

Definition 4 *A positive integer n is **deficient** if $\sigma(n) < 2n$, perfect if $\sigma(n) = 2n$, and **abundant** if $\sigma(n) > 2n$. We call a number **non-deficient** if it is abundant or perfect.*

We also formally define pseudoperfect numbers as follows.

Definition 5 *A positive integer n is a **pseudoperfect number** if there exists a subset $\mathcal{S}$ of the proper divisors of n such that*

$$n = \sum_{d \in \mathcal{S}} d.$$

The ten smallest pseudoperfect numbers are 6, 12, 18, 20, 24, 28, 30, 36, 40, and 42. The following depicts just one of the possibly several ways to express each number as a sum of proper divisors.

6	$1 + 2 + 3$
12	$2 + 4 + 6$
18	$3 + 6 + 9$
20	$1 + 4 + 5 + 10$
24	$4 + 8 + 12$
28	$1 + 2 + 4 + 7 + 14$
30	$5 + 10 + 15$
36	$6 + 12 + 18$
40	$2 + 8 + 10 + 20$
42	$7 + 14 + 21$

Table 1: A way to express the first ten pseudoperfect numbers as a sum of proper divisors

Definition 6 *For a pseudoperfect number n, a subset $\mathcal{S}$ of the proper divisors of n is a **pseudoperfect set of** n if*

$$n = \sum_{d \in \mathcal{S}} d.$$

Note that each entry in Table 1 reveals a pseudoperfect set for the respective entry. For instance, a pseudoperfect set for 24 is $\{4, 8, 12\}$. However, the table only depicts one pseudoperfect set for each integer n, and certain pseudoperfect numbers n may have several pseudoperfect sets. For example, there are two ways to express the pseudoperfect number 12 as a sum of proper divisors:

$$12 = 2 + 4 + 6 = 1 + 2 + 3 + 6.$$

And, therefore, it follows that there are 2 distinct pseudoperfect sets for 12, namely $\{2, 4, 6\}$ and $\{1, 2, 3, 6\}$.

3 Pseudoperfect Numbers

In this section, we will dive into a few of the major properties of pseudoperfect numbers that will be used in future results.

One property we can observe in Table 1 is that 6, along with each of its multiples, are pseudoperfect numbers: 12, 18, 24, 30, 36, and 42 are all pseudoperfect numbers, and one of the pseudoperfect sets of each of the multiples is a scaled up version of the pseudoperfect set of 6. The following lemma intuitively results from this observation:

Lemma 7 *If n is a pseudoperfect number, then mn is a pseudoperfect number for all positive integers m. In other words, any positive multiple of a pseudoperfect number is a pseudoperfect number.*

Proof. Let n be a pseudoperfect number. Then, there exists a pseudoperfect set of n. Suppose a pseudoperfect set of n is $\mathcal{S}_n = \{d_1, d_2, \ldots, d_k\}$. Therefore, we may write

$$n = d_1 + d_2 + \cdots + d_k$$

where

- $d_i \neq d_j$ for all $1 \leq i < j \leq k$
- $d_i \mid n$ for all $1 \leq i \leq k$, and
- $d_i < n$ for all $1 \leq i \leq k$.

It follows that

$$mn = m\left(d_1 + d_2 + \cdots + d_k\right) = md_1 + md_2 + \cdots + md_k.$$

Now, the following properties therefore must hold:

- As $d_i \neq d_j$ for all $1 \leq i < j \leq k$, it follows that $md_i \neq md_j$ for all $1 \leq i < j \leq k$.
- As $d_i \mid n$ for all $1 \leq i \leq k$, then $md_i \mid mn$ for all $1 \leq i \leq k$.
- As $d_i < n$, the $md_i < mn$. So, md_i is a proper divisor of mn.

Hence, $\mathcal{S}_{mn} = \{md_1, md_2, \ldots, md_k\}$ is a subset of proper divisors summing to mn. Therefore, mn is pseudoperfect number, as desired. ■

This idea that multiples of pseudoperfect numbers are also pseudoperfect numbers is key as it allows us to find many pseudoperfect numbers in a very quick fashion. It also gives us a technique to find pseudoperfect sets of these multiples.

Definition 8 *Let n be a pseudoperfect number where $\mathcal{S}_n = \{d_1, d_2, \ldots, d_k\}$ is a pseudoperfect set of n. For a positive integer m, we say the set*

$$S_{mn} = \{md_1, md_2, \ldots, md_k\}$$

*is a pseudoperfect set of mn inherited from n as all elements of the pseudoperfect set of mn are a scaled up version of those in the pseudoperfect set of n. We call such sets **inherited pseudoperfect sets**.*

Lemma 9 *If n is a pseudoperfect number, then $\sigma(n) \geq 2n$.*

Proof. For the sake of contradiction, suppose $\sigma(n) < 2n$. Then, the sum of the proper divisors of n is

$$\sigma(n) - n < 2n - n = n.$$

So, the sum of any subset of proper divisors of n will be at most the value of $\sigma(n) - n$, and therefore, less than n. Thus, n cannot be pseudoperfect, a contradiction. Hence, $\sigma(n) \geq 2n$. ■

Definition 10 *If n is an abundant number, but not a pseudoperfect number, n is called a **weird number**. That is, n is weird if $\sigma(n) > 2n$ but there is no subset of proper divisors of n which sum to n.*

The existence of odd weird numbers, posed by Erdös and Benkoski [1], is still an open problem. Nonetheless, we know of many weird numbers. The first six weird numbers are 70, 836, 4030, 5830, 7192, and 7912, which are all even.

Lemma 11 *There exists infinitely many weird numbers.*

Proof. We show that for any weird number n, then np is also weird for prime p satisfying the inequality $p > \sigma(n)$. Because n is weird, it is abundant, by Definition 10. Therefore, $\sigma(n) > 2n$. As $p > \sigma(n)$, then $p > n$. Also, as p is prime, it is relatively prime to n. As σ is a multiplicative function, it follows that

$$\begin{aligned} \sigma(np) &= \sigma(n)\sigma(p) \\ &= \sigma(n) \cdot (p+1) \\ &> 2n \cdot (p+1) \\ &> 2 \cdot np. \end{aligned}$$

It follows that np is abundant. We now show that np cannot be pseudoperfect if n is weird. Assume for the sake of contradiction that np is pseudoperfect. It follows that np would be expressible as the sum of a subset of its proper divisors. For every divisor in that sum, it must either be a divisor of n, or a divisor of n multiplied by p (which includes p itself). Therefore, we can write

$$np = pn = s_1 + p \cdot s_2$$

where s_1 and s_2 are each the sum of some subset of the divisors of n. Here, s_1 is the sum of all divisors which are not divisible by p while ps_2 is the sum of all divisors which are divisible by p. Therefore, it follows that

$$p(n - s_2) = s_1.$$

Hence, we have two cases:

- If $n - s_2 = 0$, then $s_2 = n$. This contradicts the fact that n is not pseudoperfect.
- If $n - s_2 > 0$, then $p \mid s_1$, meaning $p \leq s_1$. We know, however, that $s_1 \leq \sigma(n) < p$, arriving us at another contradiction.

It follows that if n is weird, then np must also be weird for primes $p > \sigma(n)$.

We then only need to show that there exists at least one weird number in order to prove that there exist infinitely many of them. Consider $n = 70$, whose proper divisors are:

$$\{1, 2, 5, 7, 10, 14, 35\}$$

Note that $\sigma(70) = 144 > 2 \cdot 70$ so 70 is abundant, but no subset of the proper divisors of 70 sums to 70, meaning it is not pseudoperfect. So, 70 is weird, and we thus have infinitely many weird numbers. ■

Note this outline gives us a good way to find infinite families of weird numbers. In particular, as $n = 70$ is a weird number and $\sigma(n) = 144$, then taking primes $p > \sigma(n) = 144$ implies np is a weird number. For example, $70 \cdot 149$, $70 \cdot 151$, and $70 \cdot 157$ are all examples of weird numbers. We can continue this process forever.

4 Constructions of Primitive Pseudoperfect Numbers

One area that continues to interest us is constructions of pseudoperfect numbers. Many papers cite the well-known construction that $2^k \cdot p$ is primitive pseudoperfect (defined and proved below). However, this is a very limited construction. Many other primitive pseudoperfect numbers take on a variety of other forms which we wished to investigate further.

Definition 12 *A pseudoperfect number n is a **primitive pseudoperfect number** if every proper divisor of n is not a pseudoperfect number.*

Lemma 13 *For a natural number k and prime p satisfying $2^k < p < 2^{k+1}$, then $2^k \cdot p$ is primitive pseudoperfect.*

Proof. Let k be a natural number and let p be an odd prime such that $2^k < p < 2^{k+1}$. Note that due to Bertrand's Postulate, for each k, there exists a prime p satisfying the necessary inequality.

First, we will show that $n = 2^k \cdot p$ is pseudoperfect. Suppose the binary representation of p is

$$p = 2^{a_1} + 2^{a_2} + \cdots + 2^{a_i},$$

for integers $0 \leq a_1 < a_2 < \cdots < a_i$.

Now, consider the following subset of divisors of $2^k \cdot p$:

$$\mathcal{S} = \left\{2^0 \cdot p,\ 2^1 \cdot p,\ 2^2 \cdot p,\ \ldots\ ,\ 2^{k-1} \cdot p,\ 2^{a_1},\ 2^{a_2},\ \ldots\ ,\ 2^{a_i}\right\}.$$

Here, $\mathcal{S}$ consists of all divisors of $2^k \cdot p$ which are divisible by p and all powers of 2 in the binary expansion of p. The sum of the elements of $\mathcal{S}$ is

$$\begin{aligned}
&2^0 \cdot p + 2^1 \cdot p + 2^2 \cdot p + \cdots + 2^{k-1} \cdot p + 2^{a_1} + 2^{a_2} + \cdots + 2^{a_i} \\
&= \left(2^0 \cdot p + 2^1 \cdot p + 2^2 \cdot p + \cdots + 2^{k-1} \cdot p\right) + (2^{a_1} + 2^{a_2} + \cdots + 2^{a_i}) \\
&= \left(2^k - 1\right) \cdot p + p \\
&= 2^k \cdot p,
\end{aligned}$$

as desired. Hence, $2^k \cdot p$ is pseudoperfect.

We now show that $n = 2^k \cdot p$ must also be primitive pseudoperfect. All proper divisors of n must be either be a divisor of 2^k (which is necessarily deficient), a divisor of p (which is also necessarily deficient), or a divisor of the form $2^j \cdot p$ for a positive integer $j < k$. In this case, since $-p < -2^k$,

$$\begin{aligned}
\sigma\left(2^j \cdot p\right) - 2 \cdot (2^j \cdot p) &= \left(2^{j+1} - 1\right) \cdot (p+1) - 2^{j+1}p \\
&= 2^{j+1}p - p + 2^{j+1} - 1 - 2^{j+1}p \\
&= 2^{j+1} - p - 1 \\
&< 2^{j+1} - 2^k - 1 \\
&\leq -1 \\
&< 0.
\end{aligned}$$

It follows that $\sigma(2^j \cdot p) < 2 \cdot (2^j \cdot p)$, which implies that $2^j \cdot p$ for $j < k$ is deficient. Thus $2^j \cdot p$ cannot be pseudoperfect for any $j < k$, so $2^k \cdot p$ has no pseudoperfect proper divisors. Therefore, n must be primitive pseudoperfect. ■

Corollary 14 *There are infinitely many primitive pseudoperfect numbers.*

Proof. It follows from Bertrand's Postulate that for all odd primes p, there exists a natural number k such that $2^k < p < 2^{k+1}$. Therefore, by Lemma 13, we are guaranteed to find a primitive pseudoperfect number for every odd prime number through this process. Because there exists infinitely many odd primes, there are infinitely many primitive pseudoperfect numbers. ■

Nicely enough this gives us a strong algorithm for finding primitive pseudoperfect numbers. Given a prime, say $p = 67$, then we may compute the integer k such that $2^k < 67 < 2^{k+1}$. It is not difficult to find $k = 6$ satisfies such a property. Therefore,

$$2^6 \cdot 67 = 4288$$

is a primitive pseudoperfect number.

While this is useful, it is not true that every primitive pseudoperfect takes on such a form. For example,

$$6790 = 2^1 \cdot 5^1 \cdot 7^1 \cdot 97^1$$

and

$$46768 = 2^4 \cdot 37^1 \cdot 79^1$$

are primitive pseudoperfect numbers which do not take on the form of $2^k p$. This motivated exploration of possible other constructions of primitive pseudoperfect numbers.

Definition 15 *A weird number n is a **primitive weird number** if every proper divisor of n is not a weird number.*

One example of a primitive weird number is 70, which we saw before in Lemma 11. In fact, it is the smallest weird number. Notably, Erdös and Benkoski [1] found 24 primitive weird numbers below 10^6.

Definition 16 *A abundant number n is a **primitive abundant number** if every proper divisor of n is not a abundant number.*

Lemma 17 *A primitive weird number n is also a primitive abundant number.*

Proof. All non-deficient numbers either perfect, pseudoperfect, or weird. We already know that any multiple of perfect or pseudoperfect numbers is also pseudoperfect, so for n to be weird, it cannot have any perfect or pseudoperfect divisor. Additionally, because n is primitive weird, it cannot have any weird proper divisor. Thus, n cannot have any non-deficient divisor, meaning that since n is necessarily abundant, it must be primitive abundant. ■

Theorem 18 *Let n be a primitive weird number. Let k be an integer such that $k < n$ and there exists a subset $\mathcal{S} = \{d_1, d_2, \ldots, d_m\}$ of proper divisors of n such that*

$$n - k = \sum_{i=1}^{m} d_i.$$

If p is a prime satisfying $\sigma(n) - 1 = kp$ and $p > n$, then np is a primitive pseudoperfect number.

Proof. Let the above conditions be satisfied. Note that we can write pn as

$$\begin{aligned} pn &= p(n-k) + pk \\ &= p\left(d_1 + d_2 + \cdots + d_m\right) + (\sigma(n) - 1). \end{aligned}$$

Let $c_1, c_2, \ldots, c_j$ be the distinct divisors of n, excluding 1. Note that

$$\sum_{i=1}^{j} c_i = \sigma(n) - 1.$$

It therefore follows that

$$\begin{aligned} pn &= p(d_1 + d_2 + \cdots + d_m) + (c_1 + c_2 + \cdots + c_j) \\ &= (pd_1 + pd_2 + \cdots + pd_m) + (c_1 + c_2 + \cdots + c_j)\,. \end{aligned}$$

Thus, we have constructed a pseudoperfect set for np. We now show np is also necessarily primitive pseudoperfect.

To do such, we show all divisors are not pseudoperfect. Consider the following three cases:

- Divisors of n. Recall n is a primitive weird number. Such divisors are immediately deficient, except for n itself which is abundant but not pseudoperfect, by Lemma 17.
- Divisors of p. Such divisors are immediately deficient.
- Divisors of the form $d \cdot p$, where d is a proper divisor of n. As n is a primitive weird number, then d is necessarily deficient implying $\sigma(d) \leq 2d - 1$.

To prove the deficiency of all $d \cdot p$, consider $2dp - \sigma(dp)$:

$$\begin{aligned} 2dp - \sigma(dp) &= 2dp - \sigma(d)\sigma(p) \\ &= 2dp - \sigma(d)(p+1) \\ &\geq 2dp - (2d-1)(p+1) \\ &= 2dp - 2dp - 2d + p + 1 \\ &= -2d + p + 1 \end{aligned}$$

We also know that $d \leq \frac{n}{2}$, and therefore that $2d \leq n$, since d is a proper divisor of n. We then have

$$\begin{aligned} 2dp - \sigma(dp) &\geq -n + p + 1 \\ &> -p + p + 1 \\ &= 1 \end{aligned}$$

Hence, $2dp > \sigma(dp)$. So, dp is deficient for all proper divisors d of n.

We conclude that no divisors of np can be pseudoperfect, and therefore that np must be necessarily primitive pseudoperfect. ■

Remark 19 *Note that per primitive weird number n, there may exist many more prime numbers p (not necessarily greater than n) such that np is primitive pseudoperfect. Because of the condition that $p > n$, primitive pseudoperfect numbers constructed using Theorem 18 tend to be exceptionally large, so this construction is difficult to use to generate primitive pseudoperfect numbers in practice.*

Theorem 20 *Let p and q be distinct primes and a be an integer such that $n = 2^a \cdot pq$. Suppose that $M = 2^{a+1} - 1$. If*

$$M^2 \leq (p - M)(q - M) \leq M^2 + M - 1,$$

then we can construct a pseudoperfect set of n. Furthermore, n is primitive pseudoperfect.

Proof. Let $x = \sigma(n) - 2n$. Then, we know that

$$\begin{aligned} x &= M \cdot (pq + p + q + 1) - 2 \cdot \left(\frac{M+1}{2}\right) \cdot pq \\ &= M \cdot (pq + p + q + 1) - (M+1) \cdot pq \\ &= M(p + q + 1) - pq \end{aligned}$$

So, if

$$M^2 \leq (p - M)(q - M) \leq M^2 + M - 1,$$

then it follows that

$$\begin{aligned} -M < (p-M)(q-M) - M^2 - M \leq -1 \\ 1 \leq M(p+q+1) - pq \leq M \\ 1 \leq x \leq M. \end{aligned}$$

Let the binary representation of x be

$$x = 2^{a_1} + 2^{a_2} + \cdots + 2^{a_k}.$$

Note that $2^{a_1}, 2^{a_2} \ldots, 2^{a_k}$ are divisors of n. So we have

$$\begin{aligned} n &= \sigma(n) - n - (2^{a_1} + 2^{a_2} + \cdots + 2^{a_k}) \\ &= \sigma(n) - n - 2^{a_1} - 2^{a_2} - \cdots - 2^{a_k}. \end{aligned}$$

Hence, n is pseudoperfect and $\text{PS}(n) = S \setminus \{2^{a_1} + \cdots + 2^{a_k}\}$, where S is the set of proper divisors of n.

We know that all the divisors of 2^a are deficient, so they are not pseudoperfect. Additionally, since $p, q \geq 2^{a+1}$, then $2^a \cdot p$ and $2^a \cdot q$ are both not pseudoperfect. So n has no pseudoperfect proper divisors. Therefore, n is primitive pseudoperfect. ■

Conjecture 21 *For all a, there exists distinct primes p and q such that*

$$M^2 \leq (p-M)(q-M) \leq M^2 + M - 1,$$

where $M = 2^{a+1} - 1$.

To demonstrate the above construction, suppose $a = 3$. Then, $M = 15$. So, we want to find distinct primes p and q such that

$$225 = 15^2 \leq (p-15)(q-15) \leq 15^2 + 15 + 1 = 241.$$

The pair of primes $p = 19$ and $q = 73$ fall in this range. This gives us $n = 2^3 \cdot 19 \cdot 73 = 11096$, which is indeed primitive pseudoperfect.
We can also construct a pseudoperfect set for 11096. We have $x = \sigma(11096) - 2 \cdot 11096 = 22200 - 2 \cdot 11096 = 8$, which gives us

$$PS(11096) = \{1, 2, 4, 8, 19, 38, 73, 76, 146, 152, 292, 584, 1387, 2774, 5548\} \setminus \{8\}$$

$$= \{1, 2, 4, 19, 38, 73, 76, 146, 152, 292, 584, 1387, 2774, 5548\}.$$

It can also be verified that the values $a = 3, p = 17, q = 131$ generate a primitive pseudoperfect number and we also found pairs of primes and corresponding primitive pseudoperfect numbers when $a = 4$.

5 Primitive Pseudoperfect Numbers

Now, we found the largest bound on

$$\frac{\sigma(n)}{n}$$

for primitive pseudoperfects n is $\frac{864}{385}$ occurring for $770 = 2 \cdot 5 \cdot 7 \cdot 11$. In particular, this is a tight bound and while this remains a conjecture at the end, we make significant progress towards such a bound.

Lemma 22 *A primitive pseudoperfect number n falls into exactly one of the three categories:*

- *n is a perfect number*
- *n is primitive abundant*
- *n has a divisor $d < n$ where $\sigma(d) \geq 2d$. In particular, n has a primitive weird divisor.*

Proof. Recall, by Lemma 9, $\sigma(n) \geq 2n$. This yields two possible cases:

- $\sigma(n) = 2n$. In this case, n is a perfect number.
- $\sigma(n) > 2n$. In this case, there are two possibilities.
  If there does not exists a divisor $d \neq n$ such that $\sigma(d) > 2d$, then n is a primitive abundant number, by Definition 17.
  Otherwise, n has a divisor $d \neq n$ such that $\sigma(d) > 2d$. In such a case, d is not a pseudoperfect number, but it is an abundant number, and so we say d is a weird number by Definition 10. There then exists a minimal weird divisor, call it d. Thus, no such divisor of d is weird, and so d is a primitive weird number, by Definition 15, as desired.

Therefore, the claim holds true. ■

Lemma 23 *If n is primitive abundant, then*

$$\frac{\sigma(n)}{n} \leq \frac{832}{385}.$$

Equality occurs only for $n = 3^2 \cdot 5 \cdot 7 \cdot 11$.

Proof. Follows from Cohen [2]. ■

Lemma 24 *If n is primitive abundant, then*

$$\frac{\sigma(n)}{n} \leq 2 + \frac{2}{s^d}$$

where s^d is the largest prime power dividing n.

Proof. Follows from Cohen [2]. ■

Theorem 25 *If n is primitive weird, then*

$$\frac{\sigma(n)}{n} \leq \frac{72}{35}.$$

Equality occurs only for $n = 2 \cdot 5 \cdot 7$.

Proof. Note that $\frac{72}{35} = 2 + \frac{2}{35}$. Let s^d be the largest prime power dividing n. We consider two cases:

- *Case 1:* Suppose $s^d \geq 35$. By Lemma 17, we know that any primitive weird number n is also a primitive abundant number. Therefore, n is a primitive abundant number and so

$$\begin{aligned} \frac{\sigma(n)}{n} &\leq 2 + \frac{2}{s^d} \\ &\leq 2 + \frac{2}{35} \\ &= \frac{72}{35}, \end{aligned}$$

  as desired. Notably, no such n, in this case, can achieve equality as 35 is not a prime power.

- *Case 2:* Suppose $s^d < 35$. The largest number attainable is

$$N = 2^5 \cdot 3^3 \cdot 5^2 \cdot 7^1 \cdot 11^1 \cdot 17^1 \cdot 19^1 \cdot 23^1 \cdot 29^1 \cdot 31^1.$$

  This is because all prime powers must be less than 35 in our product. Recall, 70 is a primitive weird number. Therefore, any other primitive weird number cannot be divisible by $70 = 2 \cdot 5 \cdot 7$. So, the largest possible primitive weird number in this case is the maximum of the following three integers:

$$\max\left(\frac{N}{2^5}, \frac{N}{5^2}, \frac{N}{7}\right) = \frac{N}{7}.$$

  Notably, this comes out to $\frac{N}{7} = 1586852229600$. The OEIS A002975 [4] sequence consists of all primitive weird numbers until 1590817681408. Therefore, as

$$1590817681408 > 1586852229600,$$

  the sequence covers all numbers we care about.
  Now, Code 66 nicely reveals that all numbers on this list have a divisor strictly greater than 31 except for 70, 836, 4030, 7192, and 45356. That is, the only primitive weird numbers where the largest prime power dividing n is strictly less than 35. Therefore, these are the only numbers we must check.

n	$\sigma(n)/n$
70	$144/70 \approx 2.057142857$
836	$1680/836 \approx 2.009569378$
4030	$8064/4030 \approx 2.000992556$
7192	$14400/7192 \approx 2.002224694$
45356	$90720/45356 \approx 2.000176382$

Table 2: $\frac{\sigma(n)}{n}$ for primitive weird number n where largest prime power is less than 35.

From here, it is clear none of the remaining values of $\frac{\sigma(n)}{n}$ strictly exceeds $\frac{144}{70} = 2 + \frac{2}{35}$, as desired.

Therefore, we may conclude that for primitive weird n, $\frac{\sigma(n)}{n} \leq \frac{72}{35}$, as desired. Equality holds when $n = 70$, as desired. ■

Theorem 26 *Let n be a primitive pseudoperfect with a smallest primitive weird divisor m. If n can be expressed as mp for a prime p where $p \nmid m$, then p is strictly larger than all primes dividing m.*

Proof. Suppose for the sake of contradiction that $q \mid m$ is some prime such that $q > p$. Consider $\frac{mp}{q} < m$. Let $m = q^k \cdot r$ where $\gcd(p, r) = \gcd(q, r) = 1$. Then,

$$\begin{aligned}
\frac{\sigma(m)/m}{\sigma(m/q)/(m/q)} &= \frac{\sigma\left(q^k\right)/q^k}{\sigma\left(q^{k-1}\right)/q^{k-1}} \\
&= \frac{\sigma\left(q^k\right)}{q \cdot \sigma\left(q^{k-1}\right)} \\
&= \frac{q^{k+1}-1}{q \cdot (q^k - 1)} \\
&= \frac{q^{k+1}-1}{q^{k+1}-q} \\
&= 1 + \frac{q-1}{q^{k+1}-q} \\
&\leq 1 + \frac{1}{q} \\
&< 1 + \frac{1}{p} \\
&= \frac{\sigma(p)}{p} \\
&= \frac{\sigma(mp/q)/(mp/q)}{\sigma(m/q)/(m/q)}
\end{aligned}$$

Therefore,

$$2 \leq \frac{\sigma(m)}{m} < \frac{\sigma\left(mp/q\right)}{mp/q}.$$

As $\frac{mp}{q}$ divides n, which is defined to be primitive pseudoperfect number, then $\frac{mp}{q}$ is not a pseudoperfect number. Thus, $\frac{mp}{q} < m$ must be a weird number. However, m is the smallest weird number, so we have a contradiction. Hence, p must be larger than all primes dividing m, as desired. ■

Theorem 27 *If n is primitive pseudoperfect and can be written as mp where p is prime and m is the smallest primitive weird divisor of n, then*

$$\frac{\sigma(n)}{n} \leq \frac{864}{385}.$$

Equality occurs only for $n = 2^1 \cdot 5^1 \cdot 7^1 \cdot 11^1$.

Proof. Firstly, assume that $p \nmid m$. If the largest prime divisor of m is at minimum 7, then by Theorem 26, it follows that $p \geq 11$. So,

$$\begin{aligned}\frac{\sigma(n)}{n} &= \frac{\sigma(mp)}{mp} \\ &= \frac{\sigma(m)}{m} \cdot \frac{\sigma(p)}{p} \\ &\leq \frac{72}{35} \cdot \frac{12}{11} \\ &= \frac{864}{385}.\end{aligned}$$

Otherwise, the largest prime divisor of m is strictly less than 7, so the set of possible prime divisors is $\{2, 3, 5\}$

- If m is divisible by 2, then $3 \nmid m$, as otherwise $6 \mid m$ and 6 is pseudoperfect so m would be pseudoperfect. The possible divisors are therefore 2 or 5. Note m cannot be divisible by 20, otherwise it would be pseudoperfect. So, either m is a power of 2 or 5 which is never abundant or m is of the form $2 \cdot 5^k$ which is also never abundant. Thus, we can ignore the case of $2 \mid n$.
- If m is not divisible by 2, then it must have divisors 3 or 5. Naturally, a prime power is deficient. Otherwise, if $m = 3^a 5^b$, then $m = 3^a 5^b$, then

$$\begin{aligned}\frac{\sigma(m)}{m} &< \frac{\sigma(3^a 5^b)}{3^a 5^b} \\ &= \frac{\left(3^{a+1} - 1\right)\left(5^{b+1} - 1\right)}{8 \cdot 3^a 5^b} \\ &< \frac{3 \cdot 5}{8} \\ &= \frac{15}{8} \\ &< 2.\end{aligned}$$

  Thus, if $p \nmid m$, the largest prime divisor of m is at most 7 and m is odd. We have just shown that no such weird number exists, so we don't need to consider this case.

On the other hand, if $p \mid m$, then suppose we write $m = p^a \cdot q$ for $\gcd(p, q) = 1$. Here, q is not necessarily prime. Note that

$$\begin{aligned}\frac{\sigma\left(p^{a+1} \cdot q\right) / \left(p^{a+1} \cdot q\right)}{\sigma\left(p^a \cdot q\right) / \left(p^a \cdot q\right)} &= \frac{\sigma\left(p^{a+1}\right)}{p \cdot \sigma\left(p^a\right)} \\ &= \frac{p^{a+2} - 1}{p^{a+2} - p} \\ &= 1 + \frac{p - 1}{p^{a+2} - p}.\end{aligned}$$

Now, as

$$\frac{\sigma\left(p^a \cdot q\right)}{\left(p^a \cdot q\right)} \leq \frac{72}{35}$$

from Theorem 25, it suffices to show

$$1 + \frac{p-1}{p^{a+2}-p} \leq \frac{12}{11} \iff \frac{p-1}{p^{a+2}-p} \leq \frac{1}{11}.$$

Notice that

$$\begin{aligned}\frac{p-1}{p^{a+2}-p} \leq \frac{1}{11} &\iff 11p - 11 \leq p^{a+2} - p \\ &\iff p^{a+2} \geq 12p - 11\end{aligned}$$

Note $a \geq 1$. So, if $p^3 \geq 12p - 11$, then it follows that

$$\begin{aligned}&(p-1) \cdot \left(p^2 + p - 11\right) \\ &= (p-1) \cdot \left(p - \left(\frac{-1-3\sqrt{5}}{2}\right)\right) \cdot \left(p - \left(\frac{-1+3\sqrt{5}}{2}\right)\right) \\ &\geq 0.\end{aligned}$$

is true for all p except $p = 2$. Hence, if $p \geq 3$, then the aforementioned value is automatically strictly less than $\frac{12}{11}$ showing the desired statement. If $p = 2$, then we have two cases:

- If $a \geq 2$, then

$$\begin{aligned}1 + \frac{p-1}{p^{a+2}-p} &= 1 + \frac{1}{2^{a+2}-2} \\ &\leq 1 + \frac{1}{2^4 - 2} \\ &= \frac{15}{14} \\ &< \frac{12}{11}\end{aligned}$$

  as desired.
- If $a = 1$, then consider $\frac{2m}{p}$ as there exists another prime divisor $p \neq 2$ dividing m. Now, rewrite $m = 2 \cdot p^k \cdot r$. Therefore, it follows that

$$\begin{aligned}\frac{\sigma(2m/p)/(2m/p)}{\sigma(m)/m} &= \frac{\sigma\left(2^2 \cdot p^{k-1} \cdot r\right) / \left(2^2 \cdot p^{k-1} \cdot r\right)}{\sigma\left(2 \cdot p^k \cdot r\right) / \left(2 \cdot p^k \cdot r\right)} \\ &= \frac{7}{6} \cdot \frac{p \cdot \sigma\left(p^{k-1}\right)}{\sigma\left(p^k\right)} \\ &= \frac{7}{6} \cdot \frac{p^{k+1} - p}{p^{k+1} - 1}\end{aligned}$$

  Hence,

$$\begin{aligned}\frac{7}{6} \cdot \frac{p^{k+1} - p}{p^{k+1} - 1} > 1 &\iff 7p^{k+1} - 7 > 6p^{k+1} - 6p \\ &\iff p^{k+1} > 7 - 6p.\end{aligned}$$

But, this is true for all p. Thus, as m is weird and therefore abundant, then $\frac{2m}{p}$ is therefore also abundant. Since n is primitive pseudoperfect and $\frac{2m}{p}$ divides n, then $\frac{2m}{p}$ is a weird number. In particular, it is a smaller weird number yielding, a contradiction as m is the smallest weird number. Thus, $a \neq 1$.

Hence,

$$1 + \frac{p-1}{p^{a+2} - p} \leq \frac{12}{11}$$

over all cases of p and a. Hence,

$$\begin{aligned} \frac{\sigma(n)}{n} &= \frac{\sigma\left(p^{a+1} \cdot q\right)}{\left(p^{a+1} \cdot q\right)} \\ &= \frac{\sigma\left(p^{a} \cdot q\right)}{\left(p^{a} \cdot q\right)} \cdot \left(1 + \frac{p-1}{p^{a+2} - p}\right) \\ &\leq \frac{72}{35} \cdot \frac{12}{11} \\ &= \frac{864}{385}, \end{aligned}$$

as desired. ■

Conjecture 28 *If n is primitive pseudoperfect,*

$$\frac{\sigma(n)}{n} \leq \frac{864}{385}.$$

Equality occurs only for $n = 2^1 \cdot 5^1 \cdot 7^1 \cdot 11^1 = 770$.

6 Pseudoperfect Sets

Now, this next section will move towards a major problem which is that there exists infinitely many n with exactly m disjoint pseudoperfect sets. To show this, we will need a few definitions before we dive into the true meat of the proof.

Definition 29 *We denote* $\text{PS}(n)$ *to be the set of pseudoperfect sets of a pseudoperfect number n.*

For example, $\text{PS}(6) = \{\{1, 2, 3\}\}$ while $\text{PS}(12) = \{\{1, 2, 3, 6\}, \{2, 4, 6\}\}$.

Definition 30 *For an odd prime p, we say $\mathcal{P}_p$ is a sequence of primes defined recursively by $p_1 = p$ and p_n is the next consecutive prime after p_{n-1} for $n \geq 2$.*

To illustrate, if $p = 3$, we can write the first 9 terms of $\mathcal{P}_p = \mathcal{P}_3$ as

$$\mathcal{P}_3 = \{3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

Definition 31 *For an odd prime p, we say $\mathcal{K}_p$ is a sequence of integers $\{k_i\}_{i \geq 1}$ defined by*

$$2^{k_i} < p_i < 2^{k_i + 1},$$

where p_i is the ith element of $\mathcal{P}_p$.

So, continuing the above example, the first 10 terms of $\mathcal{K}_3$ are

$$\{1, 2, 2, 3, 3, 4, 4, 4, 4\}$$

Definition 32 *For an odd prime p, we say $\mathcal{A}_p$ is a sequence of integers $\{a_i\}_{i\geq 1}$ defined by $a_i = 2^{k_i+1}p_i$ where p_i and k_i are the ith elements of $\mathcal{P}_p$ and $\mathcal{K}_p$, respectively.*

So, for $\mathcal{A}_3$, this sequence is

$$\{2^2 \cdot 3, 2^3 \cdot 5, 2^3 \cdot 7, 2^4 \cdot 11, 2^4 \cdot 13, 2^5 \cdot 17, 2^5 \cdot 19, 2^5 \cdot 23, 2^5 \cdot 29\}$$

Definition 33 *For an odd prime p, we say $\mathcal{N}_p$ is a sequence of integers $\{n_\ell\}_{\ell\geq 1}$ defined by*

$$\begin{aligned} n_\ell &= \prod_{j=1}^{\ell} a_j = \prod_{j=1}^{\ell} \left(2^{k_j+1} \cdot p_j\right) \\ &= \left(2^{k_1+1} \cdot p_1\right)\left(2^{k_2+1} \cdot p_2\right)\left(2^{k_3+1} \cdot p_3\right) \cdots \left(2^{k_\ell+1} \cdot p_\ell\right), \end{aligned}$$

where a_i is the ith element of $\mathcal{A}_p$ and p_i and k_i are the ith elements of $\mathcal{P}_p$ and $\mathcal{K}_p$, respectively.

Lemma 34 *For an odd prime p, let k satisfy $2^k < p < 2^{k+1}$. Then, $2^{k+1}p$ is a pseudoperfect number.*

Proof. By Lemma 13, recall $2^k p$ is a pseudoperfect number. In fact, it is a primitive pseudoperfect number. By Lemma 7, any multiple of a pseudoperfect number is a pseudoperfect number as well. Since $2^{k+1}p$ is a multiple of a pseudoperfect number, namely $2^k p$, then $2^{k+1}p$ is also pseudoperfect, as desired. ■

Lemma 35 *For an odd prime p, let $\{n_\ell\}_{\ell\geq 1}$ to be the sequence defined by $\mathcal{N}_p$. For all $\ell \geq 1$, then n_ℓ is pseudoperfect.*

Proof. We prove this via induction. Note that n_1 takes on the form of $2^{k+1}p$ where p is an odd prime and k satisfies $2^k < p < 2^{k+1}$. Hence, by Lemma 34, n_1 is pseudoperfect.

Suppose for $\ell = k$, then n_k is pseudoperfect. Now consider $\ell = k + 1$. Since n_k is pseudoperfect, by our induction hypothesis, and n_k divides n_{k+1}, it follows that n_{k+1} is pseudoperfect. Hence, as any multiple of a pseudoperfect is pseudoperfect by Lemma 7, then for all ℓ, it follows that n_ℓ is pseudoperfect, as desired. ■

Theorem 36 *Let p be an odd prime and $\{n_\ell\}_{\ell\geq 1}$ be the sequence defined by $\mathcal{N}_p$. Then,*

$$\lim_{\ell\to\infty} \frac{\sigma(n_\ell)}{n_\ell}$$

tends to infinity.

Proof. The divisor summatory function possess a nice property in that it is multiplicative. It therefore follows that

$$\begin{aligned}\frac{\sigma(n_\ell)}{n_\ell} &= \frac{\sigma\left(p_1p_2\cdots p_\ell\right)}{p_1p_2\cdots p_\ell}\cdot\frac{\sigma\left(2^{\sum k_j+\ell}\right)}{2^{\sum k_j+\ell}}\\ &= \left(\prod_{j=1}^{\ell}\frac{p_j+1}{p_j}\right)\cdot\left(\frac{2^{\sum k_j+\ell+1}-1}{2^{\sum k_j+j}}\right)\\ &= \left(\prod_{j=1}^{\ell}\frac{p_j+1}{p_j}\right)\cdot\left(2-\frac{1}{2^{\sum k_j+\ell}}\right)\end{aligned}$$

Therefore,

$$\lim_{\ell\to\infty}\frac{\sigma(n_\ell)}{n_\ell} = \lim_{\ell\to\infty}\left[\left(\prod_{j=1}^{\ell}\frac{p_j+1}{p_j}\right)\cdot\left(2-\frac{1}{2^{\sum k_j+j}}\right)\right]$$

But,

$$\lim_{\ell\to\infty}\left[\prod_{j=1}^{\ell}\left(\frac{p_j+1}{p_j}\right)\right]\to\infty$$

and

$$1-\frac{1}{2^{\sum k_j+j}}$$

is an increasing function. Thus, this entire must tend to infinity, as desired. So, overall,

$$\lim_{\ell\to\infty}\frac{\sigma(n_\ell)}{n_\ell},$$

as desired. ■

Remark 37 *A nice way to see the product of $\frac{p+1}{p}$ diverges when taking the product over all primes p is through*

$$\prod_p\frac{p+1}{p} = \prod_p\frac{1}{\left(1+\frac{1}{p}\right)^{-1}} = \frac{\prod_p\left(1-\frac{1}{p}\right)^{-1}}{\prod_p\left(1-\frac{1}{p^2}\right)^{-1}} = \frac{\zeta(1)}{\zeta(2)} = \frac{6}{\pi^2}\cdot\zeta(1).$$

As $\zeta(1)$ diverges, then the entire product diverges. Here, $\zeta(2)=\frac{\pi^2}{6}$ is often known as the Basel Problem.

Theorem 38 *Let p be an odd prime and $\{n_\ell\}_{\ell\geq 1}$ be the sequence defined by $\mathcal{N}_p$. Then,*

$$\frac{\sigma(n_\ell)}{n_\ell} < \frac{\sigma(n_{\ell+1})}{n_{\ell+1}}.$$

Proof. Recall, $n_{\ell+1} = n_\ell \cdot a_{\ell+1}$. Note that $a_\ell = 2^{k_{\ell+1}+1}p_{\ell+1}$. Hence, if $n_\ell = 2^a \cdot M$, then

$$\begin{aligned}
\frac{\sigma(n_{\ell+1})}{n_{\ell+1}} &= \frac{\sigma(2^{a+k_{\ell+1}+1} \cdot M \cdot p_{\ell+1})}{2^{a+k_{\ell+1}+1} \cdot M \cdot p_{\ell+1}} \\
&= \frac{\left(2^{a+k_{\ell+1}+2} - 1\right) \cdot \sigma(M) \cdot (p_{\ell+1}+1)}{2^{a+k_{\ell+1}+1} \cdot M \cdot p_{\ell+1}} \\
&\geq \frac{\left(2^{a+k_{\ell+1}+2} - 2^{k_{\ell+1}+1}\right) \cdot \sigma(M) \cdot (p_{\ell+1}+1)}{2^{a+k_{\ell+1}+1} \cdot M \cdot p_{\ell+1}} \\
&= \frac{(2^{a+1}-1) \cdot \sigma(M) \cdot (p_{\ell+1}+1)}{2^a \cdot M \cdot p_{\ell+1}} \\
&= \frac{\sigma\left(2^a\right) \cdot \sigma(M) \cdot (p_{\ell+1}+1)}{2^a \cdot M \cdot p_{\ell+1}} \\
&= \frac{\sigma\left(2^a \cdot M\right)}{2^a \cdot M} \cdot \frac{p_{\ell+1}+1}{p_{\ell+1}} \\
&\geq \frac{\sigma\left(2^a \cdot M\right)}{2^a \cdot M} \\
&= \frac{\sigma(n_\ell)}{n_\ell},
\end{aligned}$$

as desired. ■

Lemma 39 *Let p be an odd prime and $\{n_\ell\}_{\ell \geq 1}$ be the sequence defined by $\mathcal{N}_p$. Then,*

$$\left\lfloor \frac{\sigma(n_1)}{n_1} \right\rfloor = 2.$$

Proof. Note that n_1 takes on the form $2^{k+1}p$ where p is a prime and k is an integer satisfying $2^k < p < 2^{k+1}$. Then,

$$\begin{aligned}
\frac{\sigma\left(2^{k+1} \cdot p\right)}{2^{k+1} \cdot p} &= \frac{2^{k+2}-1}{2^{k+1}} \cdot \frac{p+1}{p} \\
&= \frac{2^{k+2} \cdot p + 2^{k+2} - p - 1}{2^{k+1} \cdot p} \\
&\geq \frac{2^{k+2} \cdot p + 2^{k+2} - \left(2^{k+1}-1\right) - 1}{2^{k+1} \cdot p} \\
&\geq \frac{2^{k+2} \cdot p + 2^{k+1}}{2^{k+1} \cdot p} \\
&= 2 + \frac{1}{p} \\
&\geq 2,
\end{aligned}$$

as desired. Now, for an upper bound,

$$\begin{aligned}\frac{\sigma\left(2^{k+1}\cdot p\right)}{2^{k+1}\cdot p} &= \frac{2^{k+2}-1}{2^{k+1}}\cdot\frac{p+1}{p}\\ &= \frac{2^{k+2}\cdot p+2^{k+2}-p-1}{2^{k+1}\cdot p}\\ &\leq \frac{2^{k+2}\cdot p+2^{k+2}}{2^{k+1}\cdot p}\\ &= 2+\frac{2}{p}\\ &= 2+\frac{2}{3}\\ &< 3.\end{aligned}$$

Therefore,

$$2\leq\frac{\sigma(n_1)}{n_1}<3 \implies \left\lfloor\frac{\sigma(n_1)}{n_1}\right\rfloor=2,$$

and so n_1 satisfies the desired property. ■

Theorem 40 *Let $p\geq 5$ be an odd prime and $\{n_\ell\}_{\ell\geq 1}$ be the sequence defined by $\mathcal{N}_p$. For every integer $k\geq 2$, there exists an integer ℓ such that*

$$\left\lfloor\frac{\sigma(n_\ell)}{n_\ell}\right\rfloor=k.$$

Proof. From Theorem 36, we know that this function tends to infinity. We must simply make sure every integer is taken on within the sequence. To do this, we use induction. Firstly, for our base case, we will show there exists an integer ℓ such that

$$\left\lfloor\frac{\sigma(n_\ell)}{n_\ell}\right\rfloor=2.$$

By Lemma 39, $\ell=1$ will satisfy such a property.

Now, for an integer $k\geq 2$, suppose we can find an integer ℓ_1 such that the claim holds. We will show that for $k+1$, we can find a corresponding integer ℓ_2. For the sake of contradiction, suppose we cannot find a corresponding ℓ_2. Then, rewrite $n_{\ell_1}=2^a\cdot M$ where M is odd. Note that M is a square-free product of primes. Then, suppose that

$$\frac{n_{\ell_1+1}}{n_{\ell_1}}=2^{j+1}p$$

If no such ℓ_2 exists, then

$$\begin{aligned}\frac{\sigma(2^a\cdot M)}{2^a\cdot M}&\leq k+1\\ k+2&\leq\frac{\sigma(2^{a+j+1}\cdot M\cdot p)}{2^{a+j+1}\cdot M\cdot p}.\end{aligned}$$

where $2^j < p < 2^{j+1}$ and $M < 2^a$ where p is prime. Note $\gcd(p_i, M) = 1$. As σ is a multiplicative function, it follows that

$$\begin{aligned}
\frac{\sigma(2^a \cdot M)}{2^a \cdot M} &\leq k+1 \\
\frac{\sigma(2^a) \cdot \sigma(M))}{2^a \cdot M} &\leq k+1 \\
\frac{(2^{a+1}-1) \cdot \sigma(M)}{2^a \cdot M} &\leq k+1 \\
\frac{\sigma(M)}{M} &\leq \frac{(k+1) \cdot 2^a}{2^{a+1}-1}.
\end{aligned}$$

In a similar fashion on the second inequality,

$$\begin{aligned}
k+2 &\leq \frac{\sigma(2^{a+j+1} \cdot M \cdot p)}{2^{a+j+1} \cdot M \cdot p} \\
k+2 &\leq \frac{\sigma(2^{a+j+1}) \cdot \sigma(M) \cdot \sigma(p)}{2^{a+j+1} \cdot M \cdot p} \\
k+2 &\leq \frac{(2^{a+j+2}-1) \cdot \sigma(M) \cdot (p+1)}{2^{a+j+1} \cdot M \cdot p} \\
\frac{(k+2) \cdot 2^{a+j+1} \cdot p}{(p+1) \cdot (2^{a+j+2}-1)} &\leq \frac{\sigma(M)}{M}.
\end{aligned}$$

Combining these two results, it follows that

$$\frac{(k+2) \cdot 2^{a+j+1} \cdot p}{(p+1) \cdot (2^{a+j+2}-1)} \leq \frac{\sigma(M)}{M} \leq \frac{(k+1) \cdot 2^a}{2^{a+1}-1}$$

Now, we will cross-multiply each side to obtain:

$$(k+2) \cdot p \cdot 2^{j+1} \cdot \left(2^{a+1}-1\right) \leq (p+1) \cdot (k+1) \cdot \left(2^{a+j+2}-1\right).$$

And now, we expand to obtain

$$\begin{aligned}
&k \cdot p \cdot 2^{a+j+2} - k \cdot p \cdot 2^{j+1} + p \cdot 2^{a+j+3} - p \cdot 2^{j+2} \leq \\
&k \cdot p \cdot 2^{a+j+2} + p \cdot 2^{a+j+2} + k \cdot 2^{a+j+2} + 2^{a+j+2} - p \cdot k - p - k - 1.
\end{aligned}$$

Simplifying, it follows that

$$\begin{aligned}
-k \cdot p \cdot 2^{j+1} + p \cdot 2^{a+j+2} - p \cdot 2^{j+2} &\leq k \cdot 2^{a+j+2} + 2^{a+j+2} - p \cdot k - p - k - 1 \\
p \cdot 2^{a+j+2} + p \cdot k + p + k + 1 &\leq k \cdot p \cdot 2^{j+1} + k \cdot 2^{a+j+2} + 2^{a+j+2} + p \cdot 2^{j+2}.
\end{aligned}$$

And, therefore, considering the left-hand side, it follows that

$$\left(p \cdot 2^{a+j+2}\right) + (p \cdot k) + (p + k + 1) > p \cdot 2^{a+j+2} + k \cdot 2^j + 2^j,$$

as $2^j < p$. Hence, substituting,

$$\begin{aligned} p \cdot 2^{a+j+2} + k \cdot 2^j + 2^j &< k \cdot p \cdot 2^{j+1} + k \cdot 2^{a+j+2} + 2^{a+j+2} + p \cdot 2^{j+2} \\ p \cdot 2^{a+2} + k + 1 &< 2kp + k \cdot 2^{a+2} + 2^{a+2} + 4p \\ 2^{a+2}(p - k - 1) &< 2kp + 4p - k - 1. \end{aligned}$$

Recall, for $k = 2$, then $p \geq 5$. So, for $k = 2$, then $2k + 1 \leq p$. Recall, for any odd prime, the next odd prime is at least two more. So, it follows that $2k+1 \leq p$ holds for all integers k and prime p. Hence, $p - k - 1 \geq k$. Thus,

$$2^{a+2} \cdot k < 2pk + 4p$$

Recall, $M < 2^a$, and so

$$4(p_1 p_2 \cdots p_{\ell_1})\, k = 4Mk < 2pk + 4p$$

Due to Bertrand's Postulate, $p_{\ell_1} < p < 2p_{\ell_1}$ as p is the next prime after p_{ℓ_1}. Hence,

$$\begin{aligned} 4(p_1 p_2 \cdots p_{\ell_1})\, k &< (2p_{\ell_1})\, k + 4 \cdot (2p_{\ell_1}) \\ 2(p_1 p_2 \cdots p_{\ell_1 - 1})\, k &< k + 4 \end{aligned}$$

But, $p_1 p_2 \cdots p_{\ell_1 - 1} \geq p_1 \geq 5$. Hence,

$$10k < k + 4 \iff 9k < 4 \iff k < \tfrac{4}{9},$$

impossible as $k \geq 2$. Thus, we obtain a contradiction. Hence, for every integer k, the claim is satisfied. ■

Theorem 41 *Let $p \geq 5$ be an odd prime and $\{n_\ell\}_{\ell \geq 1}$ be the sequence defined by $\mathcal{N}_p$. For some positive integer ℓ, suppose that*

$$\left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor = k.$$

Define $h(n_\ell) = \sigma(n_\ell) - n_\ell \cdot k$ and $s(n_\ell)$ be the sum of all positive integers which divide n_ℓ, but not a_ℓ. Then, necessarily, $s(n_\ell) \geq h(n_\ell)$.

Proof. By Theorem 40, suppose that

$$\begin{aligned} k < \frac{\sigma(n_{\ell-1})}{n_{\ell-1}} = \frac{\sigma(2^a \cdot M)}{2^a \cdot M} < k + 1 \\ k + 1 < \frac{\sigma(n_\ell)}{n_\ell} = \frac{\sigma\left(2^{a+b+1} \cdot M \cdot p\right)}{2^{a+b+1} \cdot M \cdot p} < k + 2 \end{aligned}$$

where $M < 2^a$ and $2^b < p < 2^{b+1}$. Here, we define $n_{\ell-1} = 2^a \cdot M$, $a_\ell = 2^{b+1} \cdot p$, and $n_\ell = n_{\ell-1} \cdot a_\ell$. From the first equation, it follows that

$$k \cdot 2^a \cdot M < \sigma(M) \cdot \left(2^{a+1} - 1\right) < (k+1) \cdot 2^a \cdot M. \tag{1}$$

We wish to show

$$\begin{aligned} h(n_\ell) &= \sigma\left(2^{a+b+1}\cdot M\cdot p\right) - \left(2^{a+b+1}\cdot M\cdot p\right)\cdot(k+1) \\ &< \left(2^{a+b+2}-1\right)\cdot\sigma(M) + p\cdot\left(2^{b+1}-1\right)\cdot\sigma(M) = s(n_\ell). \end{aligned}$$

Note that the right-hand side is the sum of all divisors of $2^{a+b+1}\cdot M\cdot p$ but are not divisible by $2^{b+1}\cdot p$. The first part, in particular, is the sum of all divisors of $2^{a+b+1}\cdot M$, which are the divisors not divisible by p. On the other hand, the second expression is the sum of all divisors of $2^{a+b+1}\cdot M\cdot p$ which are divisible by p, but the exponent of 2 does not exceed b. Now, this yields

$$\begin{aligned} &\left(2^{a+b+2}-1\right)\cdot\sigma(M)\cdot(p+1) - \left(2^{a+b+1}\cdot M\cdot p\right)\cdot(k+1) \\ &< \left(2^{a+b+2}-1\right)\cdot\sigma(M) + p\cdot\left(2^{b+1}-1\right)\cdot\sigma(M). \end{aligned}$$

Then, upon simplification,

$$\begin{aligned} \left(2^{a+b+2}-1\right)\cdot\sigma(M)\cdot p &< p\cdot\left(2^{b+1}-1\right)\cdot\sigma(M) + \left(2^{a+b+1}\cdot M\cdot p\right)\cdot(k+1) \\ \left(2^{a+b+2}-1\right)\cdot\sigma(M) &< \left(2^{b+1}-1\right)\cdot\sigma(M) + \left(2^{a+b+1}\cdot M\right)\cdot(k+1) \end{aligned}$$

But, due to (1), it follows that

$$\begin{aligned} &\left(2^{b+1}-1\right)\cdot\sigma(M) + \left(2^{a+b+1}\cdot M\right)\cdot(k+1) \\ &> \left(2^{b+1}-1\right)\cdot\sigma(M) + \sigma(M)\cdot\left(2^{a+1}-1\right)\cdot 2^{b+1} \end{aligned}$$

So, utilizing this information, we have that

$$\begin{aligned} &\left(2^{a+b+2}-2\right)\cdot\sigma(M) < \left(2^{b+1}-1\right)\cdot\sigma(M) + \sigma(M)\cdot\left(2^{a+1}-1\right)\cdot 2^{b+1} \\ \iff &\left(2^{a+b+2}-2\right)\cdot\sigma(M) < \left(2^{b+1}-1\right)\cdot\sigma(M) + \sigma(M)\cdot\left(2^{a+b+2}-2^{b+1}\right) \\ \iff &\left(2^{b+1}-2\right)\cdot\sigma(M) < \left(2^{b+1}-1\right)\cdot\sigma(M) \\ &\iff 0 < \sigma(M), \end{aligned}$$

which is true, as desired. ■

Theorem 42 *Let $p \geq 5$ be an odd prime and $\{n_\ell\}_{\ell\geq 1}$ be the sequence defined by $\mathcal{N}_p$. For some positive integer ℓ, suppose that*

$$\left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor = k.$$

Define $h\left(n_\ell\right) = \sigma(n_\ell) - n_\ell\cdot k$ and $\mathcal{S}$ be the set of all positive integers which divide n_ℓ, but are not divisible by a_ℓ. Then, every integer from 1 to $h(n_\ell)$ can be expressed as the sum of the elements of some subset of $\mathcal{S}$.

Proof. Previously mentioned in Theorem 41, recall that

$$h(n_\ell) = \left(2^{a+b+2}-1\right)\cdot\sigma(M) + p\cdot\left(2^{b+1}-1\right)\cdot\sigma(M).$$

To construct every integer up to $h(n_\ell)$, we will first construct all numbers up to $\left(2^{a+b+2}-1\right)\cdot\sigma(M)$ with only divisors of $2^{a+b+1}\cdot M$.

Firstly, we can construct all numbers from 1 to $2^{a+b+2}-1$. This is because each number has a unique binary representation. Taking the binary representation, we can express each number as a unique sum of powers of 2 from 2^0 until 2^{a+b+1} with no repetition.

Now, we will slowly construct all numbers from 1 until $(\sigma(M)-1)+\left(2^{a+b+2}-1\right)$. Recall M is a product of distinct odd primes, say $p_1, p_2, \ldots, p_\ell$. Then, begin a process where we add divisors of M, aside from 1, until we reach $\sigma(M)-1$:

$$0 \to p_1 \to p_1+p_1p_2 \to p_1+p_1p_2+p_2 \to \cdots$$

The order of the divisors we add doesn't matter, as long as we eventually reach $\sigma(M)$. In addition, we only add divisors one at a time. Note that each divisor is less than or equal to $M < 2^a < 2^{a+b+2}-1$. Therefore, we can construct each number in the interval of every pair of adjacent numbers using just divisors of 2^{a+b+1} and the binary representation. Hence, we can obtain the numbers at most

$$(\sigma(M)-1)+\left(2^{a+b+2}-1\right).$$

We, again, use a similar process. Add twice the divisors of M, except for $2\cdot 1$, one at a time until we eventually reach $2(\sigma(M)-1)$:

$$0 \to 2p_1 \to 2p_1+2p_1p_2 \to 2p_1+2p_1p_2+2p_2 \to \cdots$$

Each divisor is less than or equal to $2M < 2^{a+1} < 2^{a+b+2}-1+(\sigma(M)-1)$. Therefore, we can construct each number in the interval of every pair of adjacent numbers using just the divisors of 2^{a+b+2} and the divisors of M. So, we obtain the numbers at most

$$2^{a+b+2}-1+(\sigma(M)-1)+2(\sigma(M)-1) = 2^{a+b+2}-1+\left(2^2-1\right)(\sigma(M)-1).$$

We use induction to show we can construct all numbers until the desired value. Suppose we can construct all numbers from 1 until

$$(\sigma(M)-1)\left(2^k-1\right)+\left(2^{a+b+1}-1\right)$$

using the divisors of $2^{k-1}M$. We will show we can construct the numbers until

$$(\sigma(M)-1)\left(2^{k+1}-1\right)+\left(2^{a+b+1}-1\right)$$

using the divisors of 2^kM.

Suppose we are adding 2^k times each divisor of M one at a time until we eventually reach $2^k(\sigma(M)-1)$:

$$0 \to 2^kp_1 \to 2^kp_1+2^kp_1p_2 \to 2^kp_1+2^kp_1p_2+2^kp_2 \to \cdots$$

Note that each divisor is at most 2^kM, and all divisors on the aforementioned list are the divisors which divide 2^kM. Thus, the difference between any two numbers in this interval is at most 2^kM. We will therefore show

$$2^kM < 2^{a+b+2}-1+\left(2^k-1\right)(\sigma(M)-1)$$

for $k \leq a+b+1$. Now,

$$\begin{aligned} & 2^k M < 2^{a+b+2} - 1 + 2^k \cdot \sigma(M) - 2^k - \sigma(M) + 1 \\ \iff & 2^k M < 2^{a+b+2} + 2^k \cdot \sigma(M) - 2^k - \sigma(M) \\ \iff & 2^k M < 2^{a+b+2} - 2^k + \left(2^k - 1\right) \sigma(M) \end{aligned}$$

But, $M + 1 < \sigma(M)$. Thus, we show the following stronger version:

$$\begin{aligned} & 2^k M < 2^{a+b+2} - 2^k + \left(2^k - 1\right)(M+1) \\ \iff & 2^k M < 2^{a+b+2} - 2^k + 2^k M - M + 2^k - 1 \\ \iff & M + 1 < 2^{a+b+2} - 1 \end{aligned}$$

As $M < 2^a$, then $M + 1 \leq 2^a < 2^{a+b+2} - 1$ is true, as desired. ■

Theorem 43 *Let $p \geq 5$ be an odd prime and $\{n_\ell\}_{\ell \geq 1}$ be the sequence defined by $\mathcal{N}_p$. For every integer ℓ, if*

$$\left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor = k.$$

then can construct $k-1$ pairwise disjoint pseudoperfect sets of n_ℓ.

Proof. For $\ell = 1$, by Lemma 34, we know $n_1 = 2^{k+1}p$ can be used to construct one pseudoperfect set where p is prime and $2^k < p < 2^{k+1}$.

We continue by induction. Suppose the claim holds for all integers $\ell < j$. We will then show it holds for $\ell = j$. If

$$\left\lfloor \frac{\sigma(n_{\ell-1})}{n_{\ell-1}} \right\rfloor = \left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor = k,$$

then we know we construct $k-1$ pairwise disjoint pseudoperfect sets of $n_{\ell-1}$ due to the induction hypothesis. Since n_ℓ is a multiple of $n_{\ell-1}$, we scale up each of our sets to obtain $k-1$ pairwise disjoint pseudoperfect sets for n_ℓ as well.

On the other hand, suppose

$$k = \left\lfloor \frac{\sigma(n_{\ell-1})}{n_{\ell-1}} \right\rfloor \neq \left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor.$$

From our induction hypothesis, we know that there exists $k-1$ pairwise disjoint pseudoperfect sets for $n_{\ell-1}$. Therefore, when we scale up by a_ℓ, we know we can construct $k-1$ pairwise disjoint pseudoperfect sets for n_ℓ. Let $\mathcal{A}$ be the union of all elements within these $k-1$ sets. Now, by Theorem 38 and Theorem 40, it follows that

$$\left\lfloor \frac{\sigma(n_\ell)}{n_\ell} \right\rfloor = k + 1.$$

Thus, we wish to construct k pairwise disjoint psuedoperfect sets for n_ℓ. Let $\mathcal{B}$ be the set of divisors of n_ℓ. From our construction in Theorem 35, we know we can write $h(n_\ell)$ as a

sum of a subset of elements which divide n_ℓ but are not divisible by a_ℓ. Suppose the set $\mathcal{C}$ consists of a construction of numbers whose sum forms $h(n_\ell)$ and whose elements divide n_ℓ but are not divisible by a_ℓ.

Note that $\mathcal{C}$ is a subset of $\mathcal{A}\setminus\mathcal{B}$. Therefore, the elements of $(\mathcal{A}\setminus\mathcal{B})\setminus\mathcal{C}$ is a pseudoperfect set of n_ℓ. When we combine this set with our $k-1$ disjoint pseudoperfect sets, we obtain k disjoint pseudoperfect sets for n_ℓ, as desired. ■

Corollary 44 *For any natural number m, there exists infinitely many positive integers n with exactly m pairwise disjoint pseudoperfect sets.*

Proof. By Theorem 43, for every prime $p \geq 5$, we know the sequence $\mathcal{N}_p$ contains at least one element such that we can construct $k-1$ pairwise disjoint pseudoperfect sets for every integer $k \geq 2$, as desired. ■

Conjecture 45 *For all pseudoperfect numbers n such that*

$$\frac{\sigma(n)}{n} \geq 3$$

then n has at least 2 disjoint pseudoperfect sets.

There is a ton of numerical evidence to support the above conjecture, and below is just the first five pseudoperfect numbers where $\frac{\sigma(n)}{n} \geq 3$.

120	{20, 40, 60}, {1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 24, 30}
180	{30, 60, 90}, {1, 2, 3, 4, 5, 9, 10, 12, 15, 18, 20, 36, 45}
240	{40, 80, 120}, {2, 4, 6, 8, 10, 12, 16, 20, 24, 30, 48, 60}
360	{60, 120, 180}, {3, 6, 9, 12, 15, 18, 24, 30, 36, 45, 72, 90}
420	{70, 140, 210}, {5, 6, 7, 12, 15, 21, 28, 35, 42, 60, 84, 105}

7 Pseudoperfect Spectrum

The Jaccard distance, which we define below, is a way to compare the similarity of two sets. We will use this property to explore the pseudoperfect sets of certain pseudoperfect numbers. Still, there is a ton more exploration to do beyond what we have shown, particularly relating to primitive pseudoperfect numbers.

Definition 46 *Given two finite sets $\mathcal{A}$ and $\mathcal{B}$, the Jaccard index of $\mathcal{A}$ and $\mathcal{B}$ is defined by*

$$J(\mathcal{A},\mathcal{B}) = \frac{|\mathcal{A}\cap\mathcal{B}|}{|\mathcal{A}\cup\mathcal{B}|}$$

Definition 47 *Given two finite sets $\mathcal{A}$ and $\mathcal{B}$, the Jaccard distance of $\mathcal{A}$ and $\mathcal{B}$ is defined by*

$$d_J(\mathcal{A},\mathcal{B}) = 1 - J(\mathcal{A},\mathcal{B})$$

Definition 48 *A nonnegative function $g(x,y)$ describing the "distance" between neighboring points for a given set is a **metric** if*

- $g(x, y) + g(y, z) \geq g(x, z)$ *(triangle inequality)*
- $g(x, y) = g(y, x)$
- $g(x, y) \geq 0$ *and* $g(x, y) = 0$ *if and only if* $x = y$.

Lemma 49 *The Jaccard distance d_J is a metric.*

Proof. In order to prove this, we must show a few properties to be true:

1. The Jaccard distance is nonnegative. Since $|A \cup B| \leq |A \cap B|$, we have

$$1 \geq J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \text{ and}$$
$$0 \leq d_J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}.$$

   Hence, the Jaccard distance is nonnegative.

2. The Jaccard distance is symmetric. We have

$$\begin{aligned} d_J(A, B) &= 1 - \frac{|A \cap B|}{|A \cup B|} \\ &= 1 - \frac{|B \cap A|}{|B \cup A|} \\ &= d_J(B, A). \end{aligned}$$

   Hence, the Jaccard distance is symmetric.

3. The Jaccard distance satisfies $d_J(x, y) = 0$ if and only if $x = y$. We first show that $d_J(x, y) = 0$ implies $x = y$.

$$\begin{aligned} d_J(x, y) &= 0 \\ 1 - \frac{|x \cap y|}{|x \cup y|} &= 0 \\ \frac{|x \cap y|}{|x \cup y|} &= 1 \\ |x \cap y| &= |x \cup y| \\ x &= y \end{aligned}$$

   We next show that $x = y$ implies $d_J(x, y) = 0$.

$$\begin{aligned} x &= y \\ |x \cap y| = |x| = |y| &= |x \cup y| \\ \tfrac{|x \cap y|}{|x \cup y|} &= 1 \\ d_J(x, y) &= 0 \end{aligned}$$

   So, the Jaccard distance satisfies this requirement of a metric space.

4. Kosub [3] proves that the Jaccard distance satisfies the triangle inequality, which is the only non-trivial requirement of a metric in this case.

Therefore, the Jaccard distance is a metric. ■

Exercise 50 *The number* 12 *has precisely two pseudoperfect sets, namely* $\mathcal{S}_1 = \{1, 2, 3, 6\}$ *and* $\mathcal{S}_2 = \{2, 4, 6\}$. *Compute the Jaccard distance of* $\mathcal{S}_1$ *and* $\mathcal{S}_2$.

Solution 51 *Note that* $\mathcal{S}_1 \cap \mathcal{S}_2 = \{2, 6\}$ *while* $\mathcal{S}_1 \cup \mathcal{S}_2 = \{1, 2, 3, 4, 6\}$. *The Jaccard distance, therefore, is*

$$d_J(\mathcal{S}_1, \mathcal{S}_2) = 1 - J(\mathcal{S}_1, \mathcal{S}_2) = 1 - \frac{|\mathcal{S}_1 \cap \mathcal{S}_2|}{|\mathcal{S}_1 \cup \mathcal{S}_2|} = 1 - \frac{2}{5} = \boxed{\frac{3}{5}.}$$

Exercise 52 *The number* 24 *has precisely five pseudoperfect sets. Two of them are* $\mathcal{A} = \{1, 2, 3, 4, 6, 8\}$ *and* $\mathcal{B} = \{1, 3, 8, 12\}$. *Compute the Jaccard index of* $\mathcal{A}$ *and* $\mathcal{B}$.

Solution 53 *Note that* $\mathcal{A} \cap \mathcal{B} = \{1, 3, 8\}$ *while* $\mathcal{A} \cup \mathcal{B} = \{1, 2, 3, 4, 6, 8, 12\}$. *The Jaccard index, therefore, is*

$$J(\mathcal{A}, \mathcal{B}) = \frac{|\mathcal{A} \cap \mathcal{B}|}{|\mathcal{A} \cup \mathcal{B}|} = \boxed{\frac{3}{7}.}$$

Definition 54 *We define* $\text{PseuSpec}(n)$ *to be the set of all pairwise Jaccard distances between all elements in* $\text{PS}(n)$, *which recall is the set of all pseudoperfect sets of a given* n. *We refer to* $\text{PseuSpec}(n)$ *as the* ***pseudoperfect spectrum of*** n.

Definition 55 *Let* $\mathcal{A}$ *and* $\mathcal{B}$ *be pseudoperfect sets of some pseudoperfect number* k *both with the same least element* x. *We define* ***Operation abracadabra on*** $\mathcal{A}$ ***and*** $\mathcal{B}$ *as*

$$\text{abracadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$$

where the following steps are taken to create $\mathcal{A}'$ *and* $\mathcal{B}'$*:*

- *Assign* $\mathcal{A}' = \mathcal{A}$ *and* $\mathcal{B}' = \mathcal{B}$.
- *Multiply each element of* $\mathcal{A}'$ *and* $\mathcal{B}'$ *by* 6.
- *Replace the least element of* $\mathcal{A}'$ *and* $\mathcal{B}'$, *which should be* $6x$, *with* $\{x, 2x, 3x\}$.

In general, we abbreviate Operation abracadabra through the notation of (a) when necessary.

Corollary 56 *As a result, if* $\text{abracadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$, *then*

- $|\mathcal{A}| + 2 = |\mathcal{A}'|$,
- $|\mathcal{B}| + 2 = |\mathcal{B}'|$,
- $|\mathcal{A} \cap \mathcal{B}| + 2 = |\mathcal{A}' \cap \mathcal{B}'|$,
- $|\mathcal{A} \cup \mathcal{B}| + 2 = |\mathcal{A}' \cup \mathcal{B}'|$, *and*
- *if* $\mathcal{A}$ *and* $\mathcal{B}$ *are pseudoperfect sets of* k, *then* $\mathcal{A}'$ *and* $\mathcal{B}'$ *are pseudoperfect sets of* $6k$.

In the case of $k = 24$, consider the following two sets:

$$\mathcal{A} = \{1, 2, 3, 4, 6, 8\}, \text{ and}$$
$$\mathcal{B} = \{1, 3, 8, 12\}.$$

It follows that $6k = 144$, and then we can take the following steps to obtain $\mathcal{A}'$ and $\mathcal{B}'$:

$$\begin{aligned}\mathcal{A} &= \{1, 2, 3, 4, 6, 8\} \\ &\mapsto \{6, 12, 18, 24, 36, 48\} \\ &\mapsto \{1, 2, 3, 12, 18, 24, 36, 48\} = \mathcal{A}' \\ \mathcal{B} &= \{1, 3, 8, 12\} \\ &\mapsto \{6, 18, 48, 72\} \\ &\mapsto \{1, 2, 3, 18, 48, 72\} = \mathcal{B}'\end{aligned}$$

Definition 57 *Let $\mathcal{A}$ and $\mathcal{B}$ be pseudoperfect sets of some pseudoperfect number k both with the same least element x. We define* ***Operation bracadabra on $\mathcal{A}$ and $\mathcal{B}$*** *as*

$$\text{bracadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$$

where the following steps are taken to create $\mathcal{A}'$ and $\mathcal{B}'$:

- *Assign $\mathcal{A}' = \mathcal{A}$ and $\mathcal{B}' = \mathcal{B}$.*
- *Multiply each element of $\mathcal{A}'$ and $\mathcal{B}'$ by 6.*
- *Replace the least element of only $\mathcal{A}'$, which should be $6x$, with $\{x, 2x, 3x\}$.*

In general, we abbreviate Operation bracadabra through the notation of (b) when necessary.

Corollary 58 *As a result, if* $\text{bracadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$*, then*

- $|\mathcal{A}| + 2 = |\mathcal{A}'|$,
- $|\mathcal{B}| = |\mathcal{B}'|$,
- $|\mathcal{A} \cap \mathcal{B}| - 1 = |\mathcal{A}' \cap \mathcal{B}'|$,
- $|\mathcal{A} \cup \mathcal{B}| + 3 = |\mathcal{A}' \cup \mathcal{B}'|$, *and*
- *if $\mathcal{A}$ and $\mathcal{B}$ are pseudoperfect sets of k, then $\mathcal{A}'$ and $\mathcal{B}'$ are pseudoperfect sets of $6k$.*

In the case of $k = 144$, recall our sets we found previously:

$$\mathcal{A} = \{1, 2, 3, 12, 18, 24, 36, 48\}, \text{ and}$$
$$\mathcal{B} = \{1, 2, 3, 18, 48, 72\}.$$

It follows that $6k = 864$, and then we can take the following steps to obtain $\mathcal{A}'$ and $\mathcal{B}'$:

$$\begin{aligned}\mathcal{A} &= \{1, 2, 3, 12, 18, 24, 36, 48\} \\ &\mapsto \{6, 12, 18, 72, 108, 144, 216, 288\} \\ &\mapsto \{1, 2, 3, 12, 18, 72, 108, 144, 216, 288\} = \mathcal{A}' \\ \mathcal{B} &= \{1, 2, 3, 18, 48, 72\} \\ &\mapsto \{6, 12, 18, 108, 288, 432\} = \mathcal{B}'\end{aligned}$$

Definition 59 *Let $\mathcal{A}$ and $\mathcal{B}$ be pseudoperfect sets of some pseudoperfect number k each with a distinct least element. Suppose the least elements of $\mathcal{A}$ and $\mathcal{B}$ are x and y. Without the loss of generality, assume $x < y$. We define* ***Operation cadabra on $\mathcal{A}$ and $\mathcal{B}$*** *as*

$$\text{cadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$$

where the following steps are taken to create $\mathcal{A}'$ and $\mathcal{B}'$:

- *Assign $\mathcal{A}' = \mathcal{A}$ and $\mathcal{B}' = \mathcal{B}$.*
- *Multiply each element of $\mathcal{A}'$ and $\mathcal{B}'$ by 6.*
- *Replace the least element of only $\mathcal{A}'$, which should be $6x$, with $\{x, 2x, 3x\}$. Note, we only replace the set with the smaller least element.*

In general, we abbreviate Operation cadabra through the notation of (c) when necessary.

Corollary 60 *As a result, if* $\text{cadabra}(\mathcal{A}, \mathcal{B}) = (\mathcal{A}', \mathcal{B}')$*, then*

- $|\mathcal{A}| + 2 = |\mathcal{A}'|$,
- $|\mathcal{B}| = |\mathcal{B}'|$,
- $|\mathcal{A} \cap \mathcal{B}| = |\mathcal{A}' \cap \mathcal{B}'|$,
- $|\mathcal{A} \cup \mathcal{B}| + 2 = |\mathcal{A}' \cup \mathcal{B}'|$, *and*
- *if $\mathcal{A}$ and $\mathcal{B}$ are pseudoperfect sets of k, then $\mathcal{A}'$ and $\mathcal{B}'$ are pseudoperfect sets of $6k$.*

In the case of $k = 864$, consider the following two sets:

$$\mathcal{A} = \{1, 2, 3, 12, 18, 72, 108, 144, 216, 288\}, \text{ and}$$
$$\mathcal{B} = \{6, 12, 18, 108, 288, 432\}.$$

It follows that $6k = 5184$, and then we can take the following steps to obtain $\mathcal{A}'$ and $\mathcal{B}'$:

$$\begin{aligned}\mathcal{A} &= \{1, 2, 3, 12, 18, 72, 108, 144, 216, 288\} \\ &\mapsto \{6, 12, 18, 72, 108, 432, 648, 1296, 1728\} \\ &\mapsto \{1, 2, 3, 12, 18, 72, 108, 432, 648, 1296, 1728\} = \mathcal{A}' \\ \mathcal{B} &= \{6, 18, 48, 72\} \\ &\mapsto \{36, 108, 288, 432\} = \mathcal{B}'\end{aligned}$$

Theorem 61 *For any $r \in (0, 1] \cap \mathbb{Q}$, there exists a k such that $r \in \text{PseuSpec}(k)$.*

Proof. Recall, from Exercise 52, $\mathcal{A} = \{1, 2, 3, 4, 6, 8\}$ and $\mathcal{B} = \{1, 3, 8, 12\}$ are two pseudoperfect sets of 24 where

$$J(\mathcal{A}, \mathcal{B}) = \frac{3}{7}.$$

In particular,

$$|\mathcal{A} \cap \mathcal{B}| = |\{1, 3, 8\}| = 3, \text{ and}$$
$$|\mathcal{A} \cup \mathcal{B}| = |\{1, 2, 3, 4, 6, 8, 12\}| = 7.$$

From the sets $\mathcal{A}$ and $\mathcal{B}$, we can construct new pseudoperfect numbers whose $\text{PseuSpec}(k)$ contains elements in $(0, 1] \cap \mathbb{Q}$ derived from $\frac{3}{7}$.

We can look at the following chart. In each circle, given sets $\mathcal{A}$ and $\mathcal{B}$, we place the ordered pair

$$\Big(|A \cap B|,\ |A \cup B|\Big).$$

In particular, we start with $\mathcal{A} = \{1, 2, 3, 4, 6, 8\}$ and $\mathcal{B} = \{1, 3, 8, 12\}$ which corresponds to the $(3, 7)$ entry. From then on, we repeatedly perform our operations. Arrows are labeled with (a), (b), and (c) to showcase whether we perform the operation of abracadabra, bracadabra, or cadabra.

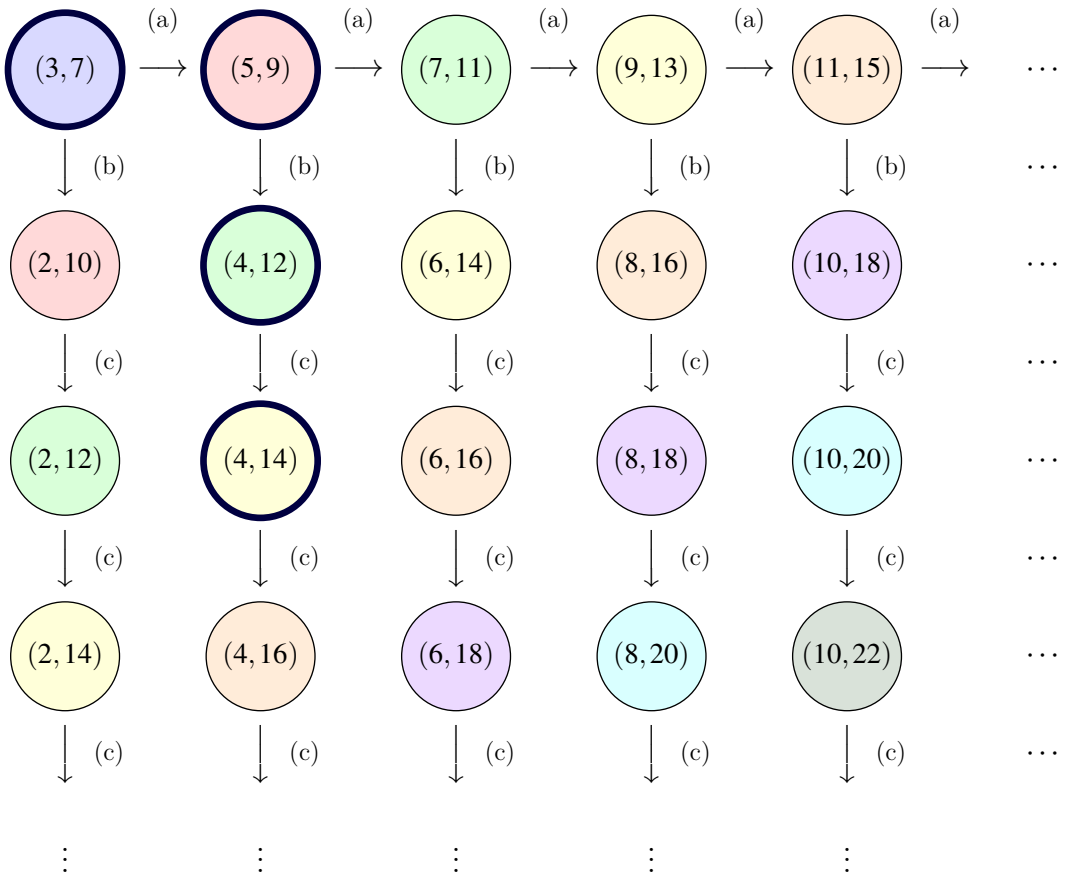


Figure 1: Graph showing the operations taken on 24 to achieve every rational in the interval $(0, 1]$.

Notably, the four red circled entries have been showcased in the examples prior. Now, for each entry (x, y) in Figure 1, we convert the value to $\frac{x}{y}$. Thus, each entry denotes the Jaccard index of the two sets, as follows:

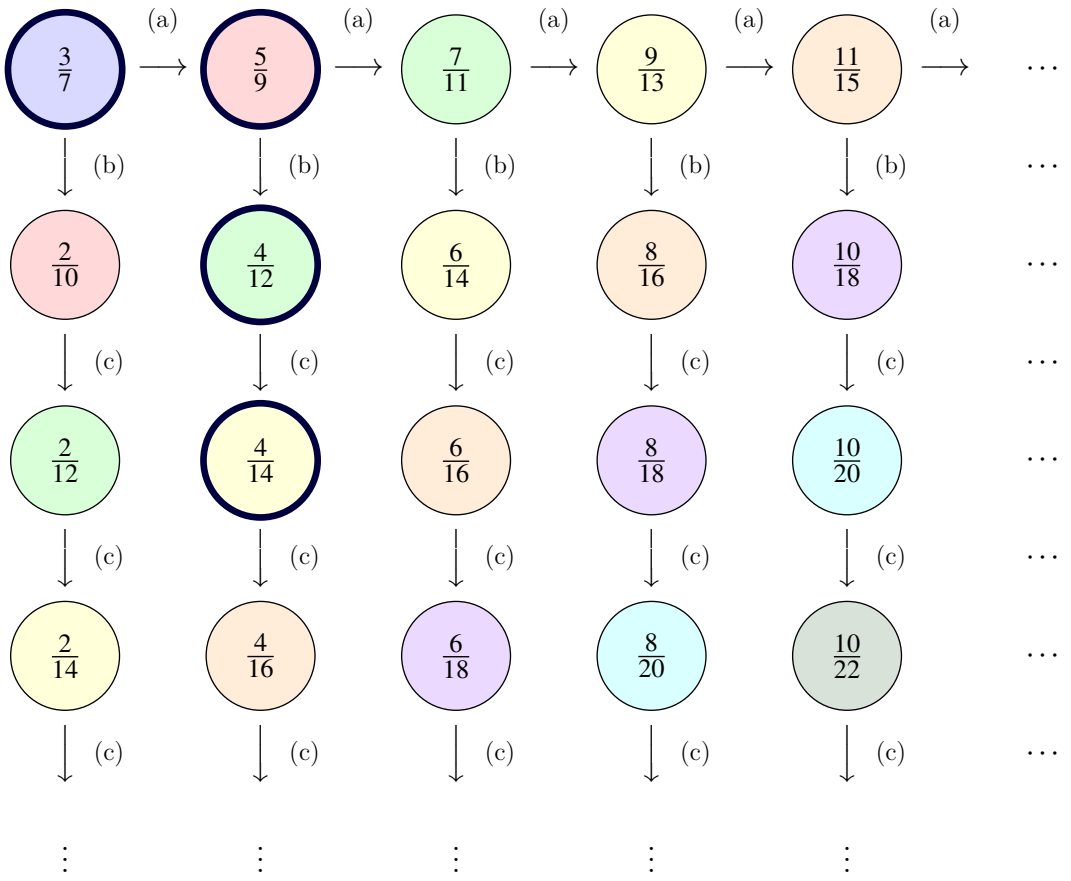


Figure 2: Graph showing the operations taken on 24 to achieve every rational in the interval $(0, 1]$.

Throughout this discussion, we will use the un-simplified fractions in Figure 2.

We want to use some patterns that naturally come out of the construction of this chart. First, note that Operation abracadabra preserves the difference between numerator and denominator, so all elements in the first row will have the same difference as $\frac{3}{7}$, namely, 4. Then, note that Operation bracadabra increases this difference by 4, so all elements of row 2 will have a difference of 8. Then, Operation cadabra increases the difference by two, so each subsequent row will have a difference of 2 more than the previous.

Additionally, because we start with the numerator 3, and Operation abracadabra increases each numerator by 2, our first row will have the odd numbers ordered as numerators. Then, because Operation bracadabra decreases the numerator by 1, the second row has all even natural numbers. Operation cadabra does nothing to change the numerator, only increasing the denominator by 2. Thus, the denominator will increase by 2 between consecutive elements in the same row, and between consecutive elements in the same column.

We claim every (reduced) fraction in the interval $(0, 1)$ appears in this table. Consider the (reduced) fraction $0 < \frac{a}{b} < 1$. Now, consider the (unreduced) $0 < \frac{8a}{8b} < 1$. Note that $8b - 8a \geq 8$ as $b - a \geq 1$. This is exactly the difference between numerator and denominator in the second row. As they were multiplied by 8, $8a$ and $8b$ are necessarily even. Thus, there exists exactly one row such that the difference between numerator and denominator is the difference between $8b$ and $8a$, and exactly one column such that the numerator is $8a$. Hence, this fraction will eventually appear in the sequence, as desired.

Thus, every rational possible value of $J(\mathcal{A}, \mathcal{B})$ strictly between 0 and 1 appears. Therefore, every value of $d_J(\mathcal{A}, \mathcal{B})$ strictly between 0 and 1 appears as $J(\mathcal{A}, \mathcal{B}) + d_J(\mathcal{A}, \mathcal{B}) = 1$.

For the value of 1, which would not appear in the above table, consider $n = 120$ where

$$\mathcal{A} = \{20, 40, 60\}$$
$$\mathcal{B} = \{1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 24, 30\}$$

Then,

$$d_J(\mathcal{A}, \mathcal{B}) = 1 - \frac{|\mathcal{A} \cap \mathcal{B}|}{|\mathcal{A} \cup \mathcal{B}|} = 1 - \frac{0}{15} = 1.$$

Therefore, we can construct every rational in the interval $(0, 1]$, as desired. ■

Remark 62 *For instance, if we wished to construct $\frac{1}{3}$, then*

$$\frac{8}{24} \leftarrow \frac{8}{22} \leftarrow \frac{8}{20} \leftarrow \frac{8}{18} \leftarrow \frac{8}{16} \leftarrow \frac{9}{13} \leftarrow \frac{7}{11} \leftarrow \frac{5}{9} \leftarrow \frac{3}{7}.$$

And so, we can trace our steps back (each time, we multiply by 6) until we get back our original fraction.

Corollary 63 *Indeed, this also implies we can get each fraction infinitely many times with infinitely many n as*

$$\frac{1}{3} = \frac{8}{24} = \frac{16}{48} = \frac{24}{72} = \cdots$$

This includes $r = 1$ which follows from previous theorems.

Remark 64 *It is interesting to note that in figure 2, the rational numbers on the same colored diagonal correspond to the same integer. On the first diagonal, colored blue, $\frac{3}{7}$ appears. The second diagonal, colored red, contains $\frac{5}{9}$ and $\frac{2}{10}$, which both correspond to $24 \cdot 6$. Every diagonal starting in the nth place to the right corresponds to a subset of the pseudoperfect spectrum of $24 \cdot 6^{n-1}$. In fact, the entire top-left $n \times n$ triangle is a subset of the pseudoperfect spectrum of $24 \cdot 6^{n-1}$, since multiplying two pseudoperfect sets by the same number keeps the Jaccard index and distance. Thus, we can say that the size of* PseuSpec(n) *can be arbitrarily large.*

Remark 65 *It is also significant that the initial $n = 24$ and multiplication by 6 are convenient choices, but analogous operations could be defined and different orders of rational numbers could be constructed from different starting values of n and different choices of perfect, or even pseudoperfect numbers to multiply by.*

8 Future Directions

Interesting extensions of this project could include generalizing the bound for primitive pseudoperfect numbers, as well as investigating the possibility of an upper bound on the Jaccard distances of primitive pseudoperfect numbers. Finding more reliable constructions for primitive pseudoperfect numbers could aid in this process and give insight into PseuSpec(n) for n primitive pseudoperfect.

Additional future directions could be exploring conditions on n that determine whether or not it has a non-inherited pseudoperfect set or, similarly, a pseudoperfect set containing the number 1. There are some numbers that can be easily checked to **not** have this property, such as those of the form $2 \cdot 3 \cdot p$ for some prime $p > 12$. We believe there to be far more such numbers, and that they fall into a more generalizable class.

We explored the topic of disjoint pseudoperfect sets, but believe there is more to be discovered in this area, including the conjectured bound for n to have disjoint pseudoperfect sets.

Finally, analogues of Theorem 61 and its related operations may give more structure to PseuSpec(n) for pseudoperfect number n, not primitive pseudoperfect, including answers to questions along the lines of how how big does a number n have to be to contain a given $r \in (0, 1] \cap \mathbb{Q}$ in PseuSpec(n).

9 Conclusion

Throughout our project, we investigated primitive pseudoperfect numbers and pseudoperfect sets.

Regarding primitive pseudoperfect numbers, we were challenged by the lack of reliable constructions to generate them, as they do not have the properties of pseudoperfect numbers that allow us to construct infinite families. The most reliable construction we are aware of is $2^k p$ where $2^k < p < 2^{k+1}$. We were able to create other constructions for primitive pseudoperfect numbers, albeit ones with conditions far more difficult to satisfy. We also constructed a bound on specific classes of primitive pseudoperfect numbers that is very likely generalizable.

In addition, we investigated properties of pseudoperfect sets, finding a major regarding the existence of disjoint pseudoperfect sets for a number: for any natural number, there exists some pseudoperfect number with exactly that many disjoint pseudoperfect sets. We also examined the set PseuSpec(n), and explored a key property that every rational number between 0 and 1 can be obtained in the pseudoperfect spectrum for some n.

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11 Appendix

Code 66 *The following code, written in C++, was used specifically for the OEIS A002975 [4] sequence. We can paste all* 1161 *entries of the table linked and it will output all values which are divisible by primes from* 2 *to* 31*. In particular, if it has any prime power greater than* 35*, then we ignore the number. It will print out just* 70*,* 836*,* 4030*,* 7192*, and* 45356*.*

The algorithm is to simply divide each number by all primes from 2 *to* 31 *until it is no longer divisible by any prime in that range. If the value* 1 *remains, then we know it is only divisible by primes from* 2 *to* 31*, and we output the number. Otherwise, there will exist a prime greater than* 35 *that divides the given number.*

```
#include <bits/stdc++.h>
using namespace std;

// Paste the entries of
// https://oeis.org/A002975/b002975.txt
// as testcases

int main() {
  int t; cin >> t;
    while (t--) {
        int idx, a; cin >> idx >> a;
        int num = a;

        while (a % 2 == 0) {
            a /= 2;
        }
        while (a % 3 == 0) {
            a /= 3;
        }
        while (a % 5 == 0) {
            a /= 5;
        }
        while (a % 7 == 0) {
            a /= 7;
        }
        while (a % 11 == 0) {
            a /= 11;
        }
        while (a % 13 == 0) {
            a /= 13;
        }
        while (a % 17 == 0) {
            a /= 17;
        }
        while (a % 19 == 0) {
            a /= 19;
        }
        while (a % 23 == 0) {
            a /= 23;
        }
        while (a % 29 == 0) {
            a /= 29;
        }
```

```
        while (a % 31 == 0) {
            a /= 31;
        }

        if (a == 1) {
            cout << num << endl;
        }
    }
}
```

Code 67 *This code computes all primitive pseudoperfects. One can adjust the bounds to reach more. To do this, we used dynamic programming to store the number of times each number can be created from some subset. This method is actually particularly nice as it finds the number of times a certain number can be formed by printing out* $\mathrm{dp}[i]$. *We adapted this code in a variety of other ways, for example, to find patterns relating to whether* $\frac{\sigma(n)}{n}$ *is greater than a specific constant* C *and patterns in odd primitive pseudoperfect numbers.*

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector <int> primitive;
    for (int i = 6; i < 60000; i++) {

        vector <int> divisors;
        for (int j = 1; j*j <= i; j++) {
            if (i % j == 0) {
                divisors.push_back(j);
                if (j*j != i) {
                    divisors.push_back(i / j);
                }
            }
        }

        sort(divisors.begin(), divisors.end());

        vector <int> dp(i+1);
        dp[0] = 1; // 1 way to get 0
        for (int j = 0; j < divisors.size(); j++) {
            for (int k = i; k >= 0; k--) {
                if (divisors[j] + k <= i) {
                    dp[divisors[j] + k] += dp[k];
                }
            }
        }

        if (dp[i] > 1) {
            int checker = 0;
            for (int j = 0; j < primitive.size(); j++) {
                if (i % primitive[j] == 0) {
                    checker = 1;
                }
            }

```

```
            if (checker == 0) {
                cout << i << endl;
                primitive.push_back(i);
            }
        }
    }
}
```

Data 68 *The following table details all the pseudoperfect number where* $\frac{\sigma(n)}{n} \geq 3$. *The leftmost column indicates the pseudoperfect number* n. *The rightmost column indicates the prime factorization. The middle column indicates* $\sigma(n) - 3n$.

Interestingly, 30240 *and* 32760 *both satisfy* $\sigma(n) = 3n$. *We believe this is particularly interesting as if* $\sigma(n) = kn$ *where* n *is pseudoperfect and there exists* $k - 1$ *disjoint pseudoperfect sets and there exists a multiple of* n, *say* mn, *where* $\sigma(mn) = (k+1)mn$ *then* mn *has* k *disjoint pseudoperfect sets.*

If we can find a mn, *then we can consistently scale up to form infinitely many pseudoperfect numbers with* k *disjoint pseudoperfect sets, was our initial approach in Section 6. Unfortunately, the existence of infinitely many multiply perfect numbers are unknown.*

27720	1440	$2^3 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 11^1$
30240	0	$2^5 \cdot 3^3 \cdot 5^1 \cdot 7^1$
32760	0	$2^3 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 13^1$
50400	1512	$2^5 \cdot 3^2 \cdot 5^2 \cdot 7^1$
55440	10368	$2^4 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 11^1$
60480	1920	$2^6 \cdot 3^3 \cdot 5^1 \cdot 7^1$
65520	8736	$2^4 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 13^1$
75600	5120	$2^4 \cdot 3^3 \cdot 5^2 \cdot 7^1$
83160	12960	$2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \cdot 11^1$
85680	5472	$2^4 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 17^1$
90720	3024	$2^5 \cdot 3^4 \cdot 5^1 \cdot 7^1$
95760	3840	$2^4 \cdot 3^2 \cdot 5^1 \cdot 7^1 \cdot 19^1$
98280	10080	$2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \cdot 13^1$
100800	6248	$2^6 \cdot 3^2 \cdot 5^2 \cdot 7^1$

Code 69 *The following code computes the prime factorization of a particular number.*

```
#include <bits/stdc++.h>
using namespace std;

int main() {
  int t; cin >> t;
    while (t--) {
        int n;
        cin >> n;
        int cntt = 0;
        for (int i = 2; i <= sqrt(n); i++) {
            int cnt = 0, checker = 0;
            while (n % i == 0) {
                checker = 1;
```

```
                n /= i;
                cnt += 1;
            }
            if (checker == 1) {
                if (cntt == 1) {
                    cout << " * ";
                }
                cntt = 1;
                cout << i << "^" << cnt;
            }
        }

        if (n != 1) {
            if (cntt == 1) {
                cout << " * ";
            }
            cout << n << "^" << 1;
        }

        cout << endl;
    }
}
```

Code 70 *The first code, similar to Code 71, computes the pseudoperfect spectrum and all pseudoperfect sets for a primitive pseudoperfect number.*

```
primitive = []
for n in range(1, 10000):
    pseudoperfect = False
    divisors_set = divisors(n)
    divisors_subsets = list(Subsets(divisors_set))
    PseudoSpec = []
    for subset in divisors_subsets:
        if (sum(subset) == n) and (n not in subset):
            pseudoperfect = True
            PseudoSpec.append(subset)
    attainablevals = []
    if pseudoperfect == True:
        checkerr = 0
        for e in primitive:
            if (n % e == 0):
                checkerr = 1
        if checkerr == 0:
            primitive.append(n)
            for set1 in PseudoSpec:
                for set2 in PseudoSpec:
                    if (set1 != set2):
                        num1 = len(set1.union(set2))
                        num2 = len(set1.intersection(set2))
                        ans = num2 / num1
                        checker = 1
                        for e in attainablevals:
                            if (e == ans):
                                checker = 2
                        if (checker == 1):
```

```
                                attainablevals.append(ans)
            print(f"PseudoSpec for {n}: {PseudoSpec}")
            attainablevals = sorted(attainablevals)
            print(f"PseuSpec for {n}: {[QQ(x) for x in attainablevals]}
    ")
            print(len(attainablevals))
```

Code 71 *This second code is very similar to Code 70, but instead, it computes the pseudoperfect spectrum and all pseudoperfect sets for a pseudoperfect number.*

```
primitive = []
for n in range(1, 10000):
    pseudoperfect = False
    divisors_set = divisors(n)
    divisors_subsets = list(Subsets(divisors_set))
    PseudoSpec = []
    for subset in divisors_subsets:
        if (sum(subset) == n) and (n not in subset):
            pseudoperfect = True
            PseudoSpec.append(subset)
    attainablevals = []
    if pseudoperfect == True:
        checkerr = 0
        for e in primitive:
            if (n % e == 0):
                checkerr = 1
        if checkerr == 0 or checkerr == 1:
            primitive.append(n)
            for set1 in PseudoSpec:
                for set2 in PseudoSpec:
                    if (set1 != set2):
                        num1 = len(set1.union(set2))
                        num2 = len(set1.intersection(set2))
                        ans = num2 / num1
                        checker = 1
                        for e in attainablevals:
                            if (e == ans):
                                checker = 2
                        if (checker == 1):
                            attainablevals.append(ans)
            print(f"PseudoSpec for {n}: {PseudoSpec}")
            attainablevals = sorted(attainablevals)
            print(f"PseuSpec for {n}: {[QQ(x) for x in attainablevals]}
    ")
            print(len(attainablevals))
```

Data 72 *The following table displays all elements in the pseudoperfect spectrum for all pseudoperfects until* 80.

6	
12	$\frac{2}{5}$
18	$\frac{2}{5}$
20	
24	$\frac{1}{7}, \frac{2}{7}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}, \frac{4}{7}$
28	
30	$\frac{2}{7}, \frac{1}{3}, \frac{2}{5}$
36	$\frac{1}{7}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}, \frac{4}{7}$
40	$\frac{2}{7}, \frac{3}{7}, \frac{4}{7}$
42	$\frac{2}{5}$
48	$\frac{1}{8}, \frac{1}{7}, \frac{2}{9}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{1}{2}, \frac{5}{9}, \frac{4}{7}, \frac{5}{8}$
54	$\frac{1}{3}, \frac{2}{5}, \frac{1}{2}$
56	$\frac{4}{7}$
60	$\frac{1}{11}, \frac{1}{10}, \frac{1}{9}, \frac{1}{8}, \frac{1}{7}, \frac{1}{6}, \frac{2}{11}, \frac{1}{5}, \frac{2}{9}, \frac{1}{4}, \frac{3}{11}, \frac{2}{7}, \frac{3}{10}, \frac{1}{3}, \frac{4}{11}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{1}{2}, \frac{5}{9}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}$
66	$\frac{1}{3}$
72	$\frac{1}{10}, \frac{1}{9}, \frac{1}{8}, \frac{1}{7}, \frac{2}{11}, \frac{1}{5}, \frac{2}{9}, \frac{1}{4}, \frac{3}{11}, \frac{2}{7}, \frac{3}{10}, \frac{1}{3}, \frac{4}{11}, \frac{3}{8}, \frac{2}{5}, \frac{3}{7}, \frac{4}{9}, \frac{5}{11}, \frac{1}{2}, \frac{5}{9}, \frac{4}{7}, \frac{3}{5}, \frac{5}{8}, \frac{2}{3}$
78	
80	$\frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{3}{8}, \frac{3}{7}, \frac{4}{9}, \frac{1}{2}, \frac{4}{7}, \frac{5}{8}$

Table 3: Pseudoperfect Spectrum for Pseudoperfects at most 80.

Code 73 *The code does essentially the same thing as the prior code. It also adds the Jaccard distance for each pair of pseudoperfect sets.*

```
for n in range(1, 100):
    pseudoperfect = False
    divisors_set = divisors(n)
    divisors_subsets = list(Subsets(divisors_set))
    PseudoSpec = []
    for subset in divisors_subsets:
        if (sum(subset) == n) and (n not in subset):
            pseudoperfect = True
            PseudoSpec.append(subset)
    if pseudoperfect == True:
        print(f"PseudoSpec for {n}: {PseudoSpec}")
        for pair in Combinations(PseudoSpec, 2).list():
            Jaccard = len(Set(pair[0]).intersection(Set(pair[1]))) /
   len(Set(pair[0]).union(Set(pair[1])))
            distance = 1 - Jaccard
            if distance > 0:
                print(f"Jaccard distance between {pair[0]} and {pair
   [1]}: {distance}")
    print("\n")
```

References

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[3] S. Kosub. A note on the triangle inequality for the jaccard distance. *Pattern Recognition Letters*, 2016.

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